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Iyoshi et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **REFRIGERATION CYCLE SYSTEM**

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Dec. 1, 2020 (JP) 2020-199794

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F25B 47/02 (2006.01)
F25B 9/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***F25B 47/025*** (2013.01); ***F25B 9/008*** (2013.01); ***F25B 9/10*** (2013.01); ***F25B 13/00*** (2013.01);
(Continued)

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CPC F25B 47/025; F25B 9/008; F25B 9/10; F25B 13/00; F25B 2313/007; F25B 2313/0233
See application file for complete search history.

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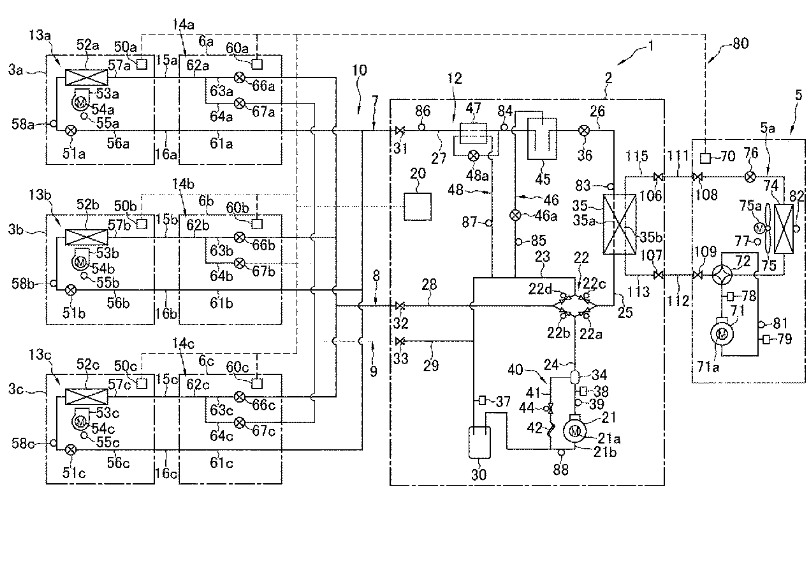
(57) **ABSTRACT**

A primary refrigerant circuit allows circulation of a primary refrigerant and includes a primary compressor, a cascade heat exchanger, a primary heat exchanger, and a primary switching mechanism. A secondary refrigerant circuit allows circulation of a secondary refrigerant and includes a secondary compressor, the cascade heat exchanger, a utilization heat exchanger, and a secondary switching mechanism. The secondary refrigerant circuit includes a bypass flow path connecting a portion between the utilization heat exchanger and the cascade heat exchanger and a suction flow path of the secondary compressor, and a bypass expansion valve provided on the bypass flow path. Executed is defrosting operation of circulating the primary refrigerant in the order of the primary compressor, the primary heat exchanger, and the cascade heat exchanger, and circulating the second refrigerant in the order of the secondary compressor, the cascade heat exchanger, and the bypass flow path.

20 Claims, 9 Drawing Sheets



(51) **Int. Cl.**
F25B 9/10 (2006.01)
F25B 13/00 (2006.01)

(52) **U.S. Cl.**
CPC . *F25B 2313/007* (2013.01); *F25B 2313/0233* (2013.01)

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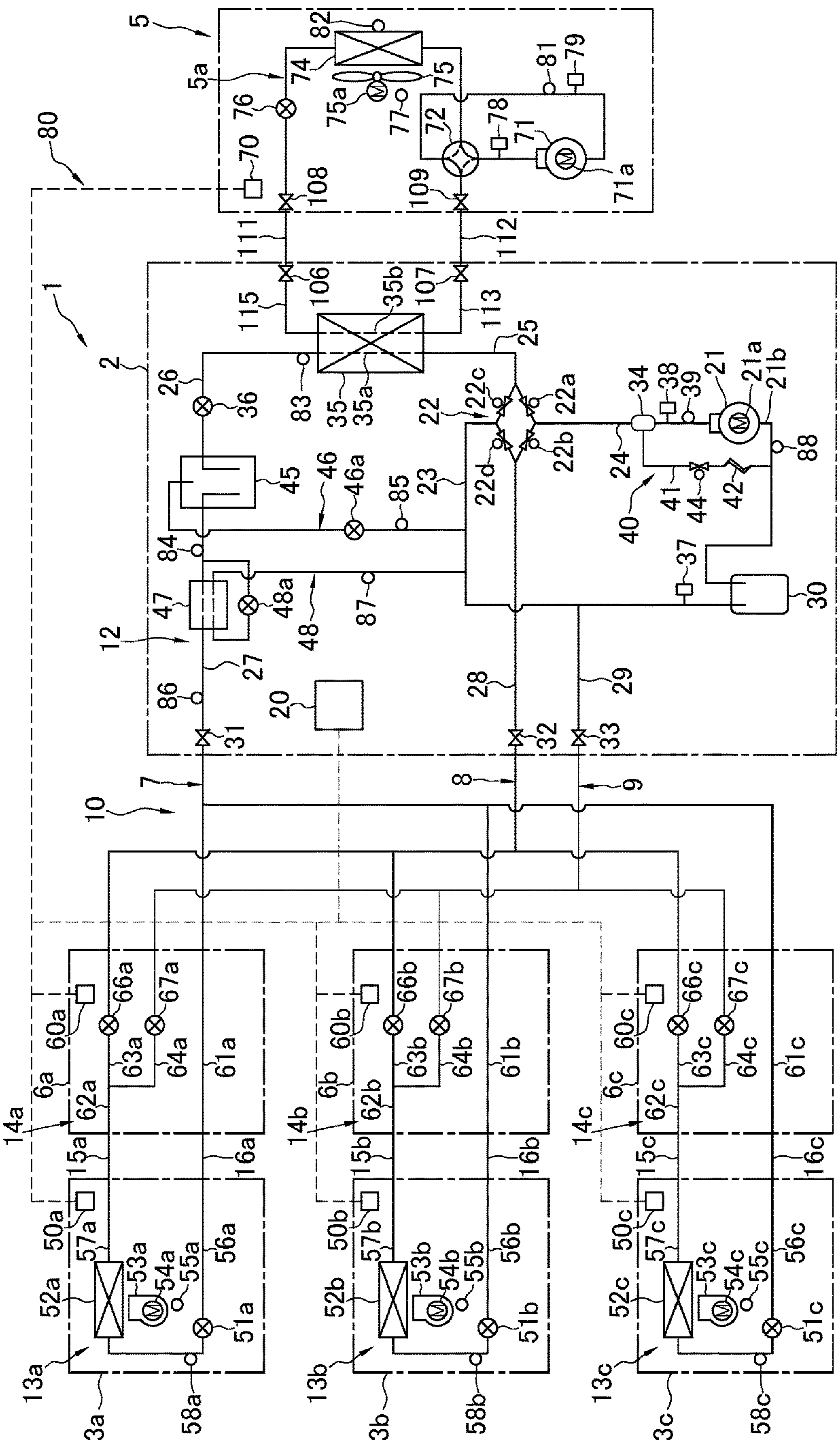
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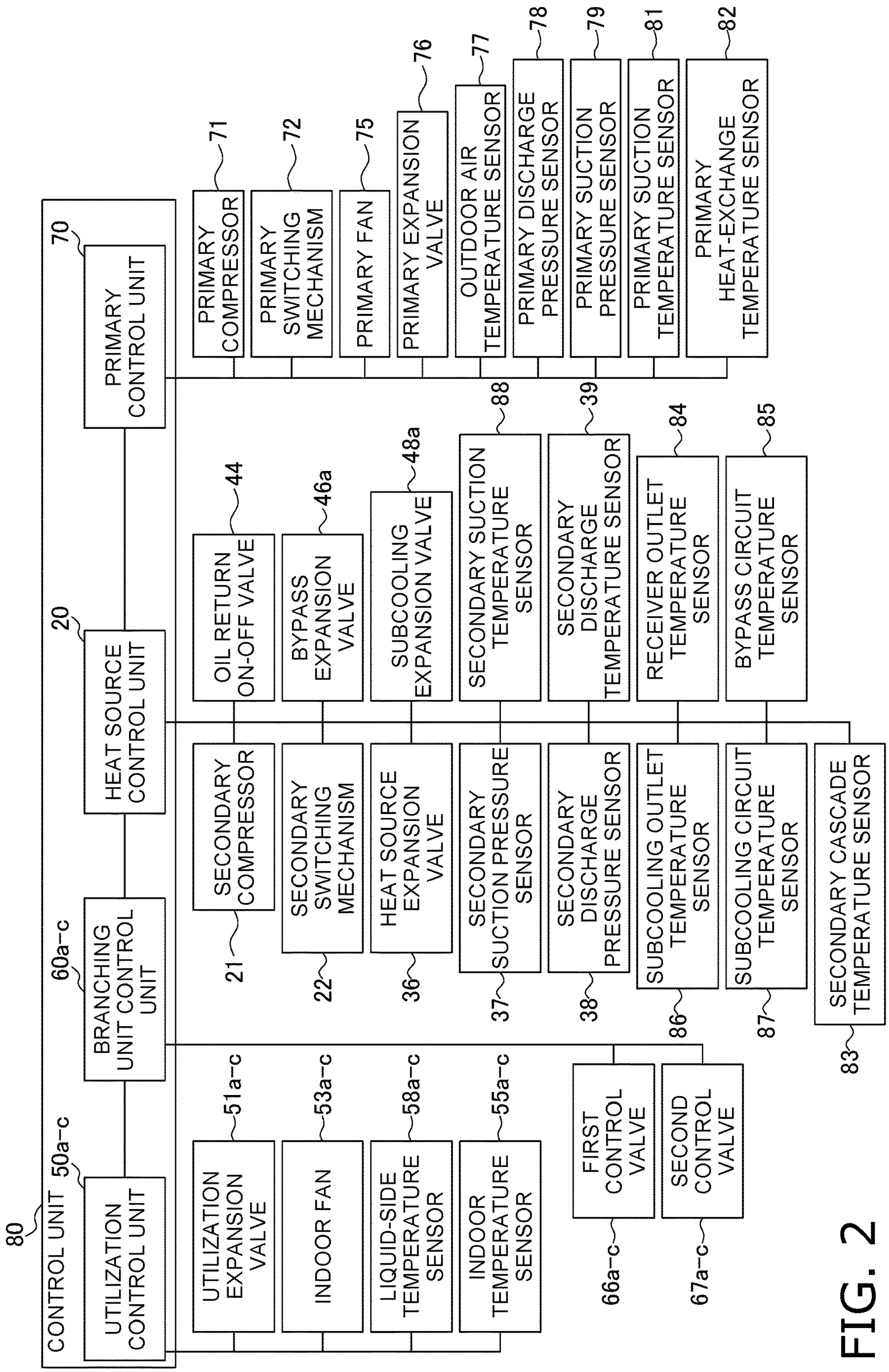
58c

FIG. 1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CONTROL UNIT
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UTILIZATION CONTROL UNIT
60a-c
BRANCHING UNIT CONTROL UNIT
20
HEAT SOURCE CONTROL UNIT
70
PRIMARY CONTROL UNIT
UTILIZATION EXPANSION VALVE
51a-c
INDOOR FAN
53a-c
LIQUID-SIDE TEMPERATURE SENSOR
58a-c
INDOOR TEMPERATURE SENSOR
55a-c
FIRST CONTROL VALVE
66a-c
SECOND CONTROL VALVE
67a-c
SECONDARY COMPRESSOR
21
SECONDARY SWITCHING MECHANISM
22
HEAT SOURCE EXPANSION VALVE
36
SECONDARY SUCTION PRESSURE SENSOR
37
SECONDARY DISCHARGE PRESSURE SENSOR
38
SUBCOOLING OUTLET TEMPERATURE SENSOR
86
SUBCOOLING CIRCUIT TEMPERATURE SENSOR
87
SECONDARY CASCADE TEMPERATURE SENSOR
83
OIL RETURN ON-OFF VALVE
44
BYPASS EXPANSION VALVE
46a
SUBCOOLING EXPANSION VALVE
48a
SECONDARY SUCTION TEMPERATURE SENSOR
88
SECONDARY DISCHARGE TEMPERATURE SENSOR
39
RECEIVER OUTLET TEMPERATURE SENSOR
84
BYPASS CIRCUIT TEMPERATURE SENSOR
85
PRIMARY COMPRESSOR
71
PRIMARY SWITCHING MECHANISM
72
PRIMARY FAN
75
PRIMARY EXPANSION VALVE
76
OUTDOOR AIR TEMPERATURE SENSOR
77
PRIMARY DISCHARGE PRESSURE SENSOR
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PRIMARY SUCTION PRESSURE SENSOR
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PRIMARY SUCTION TEMPERATURE SENSOR
81
PRIMARY HEAT-EXCHANGE TEMPERATURE SENSOR
82

FIG. 2

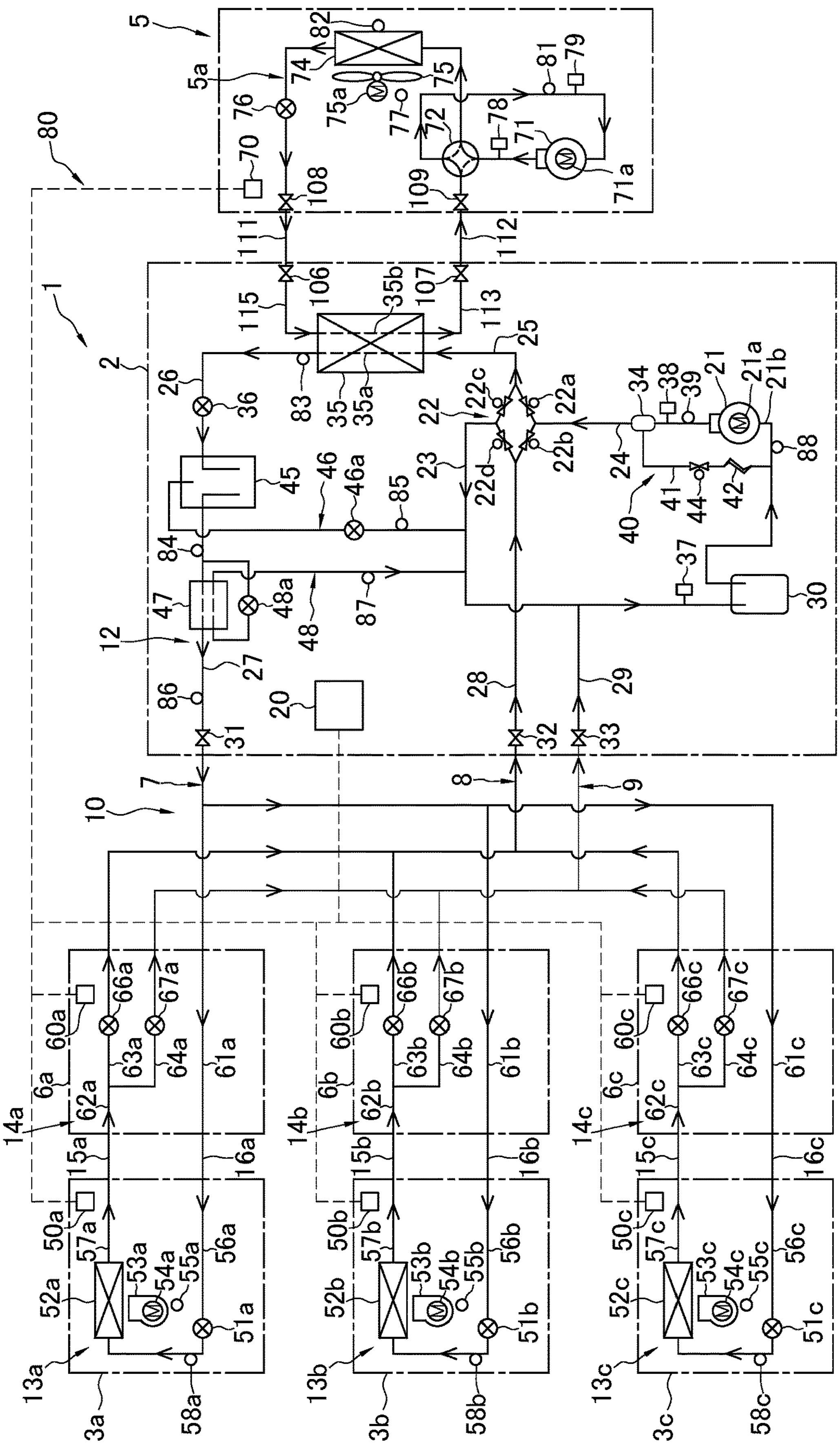
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FIG. 3

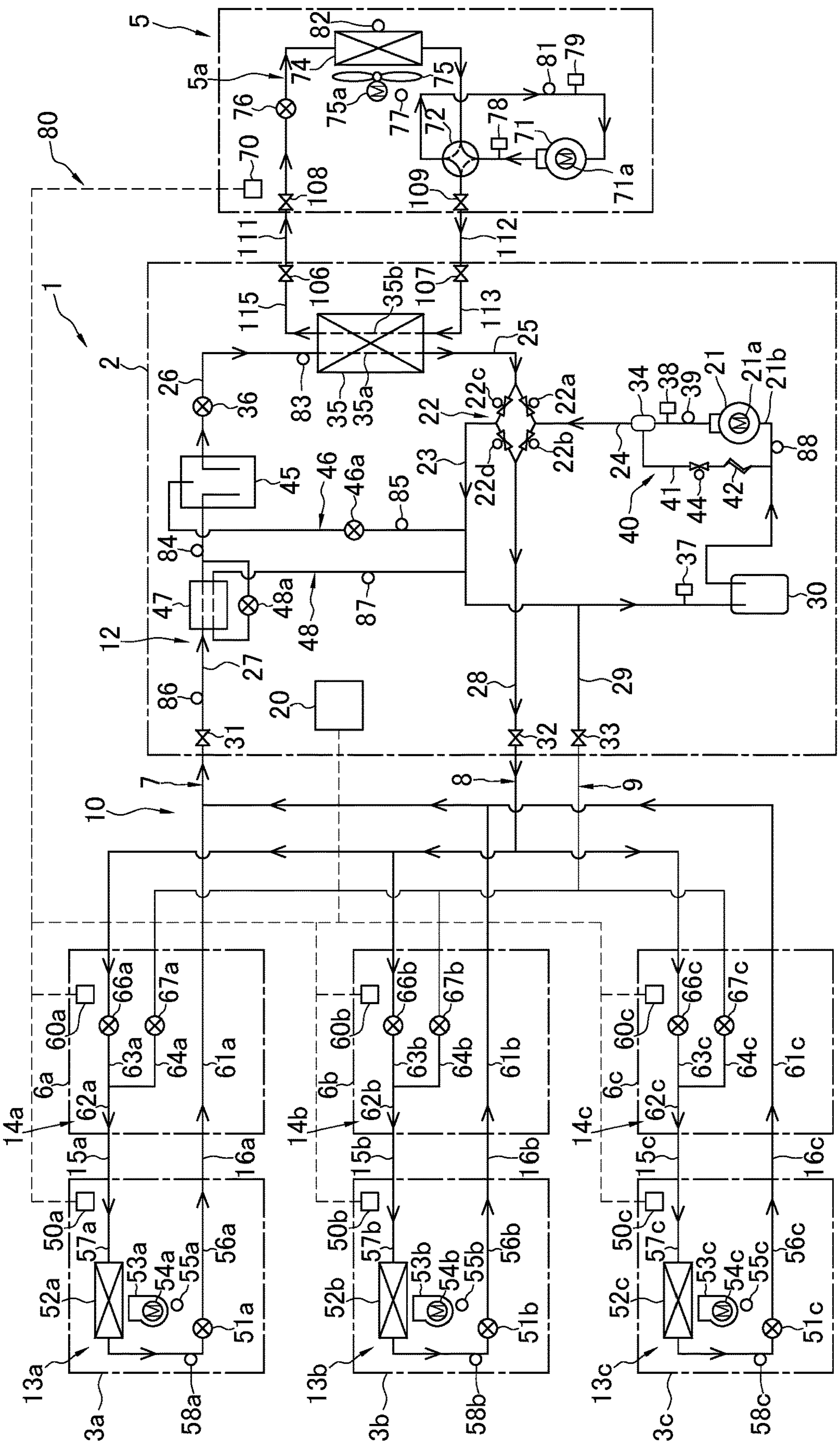

FIG. 4

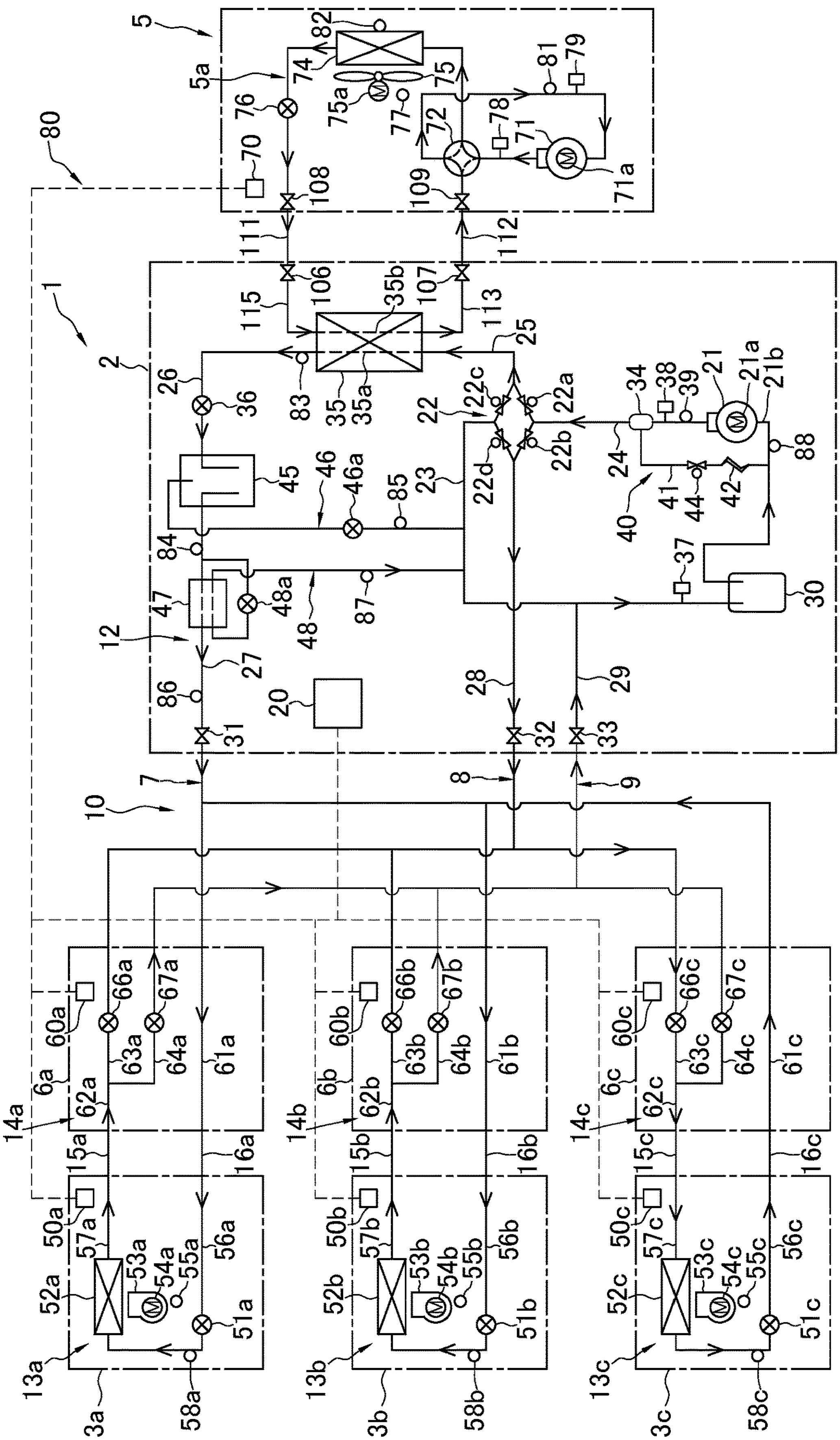

FIG. 5

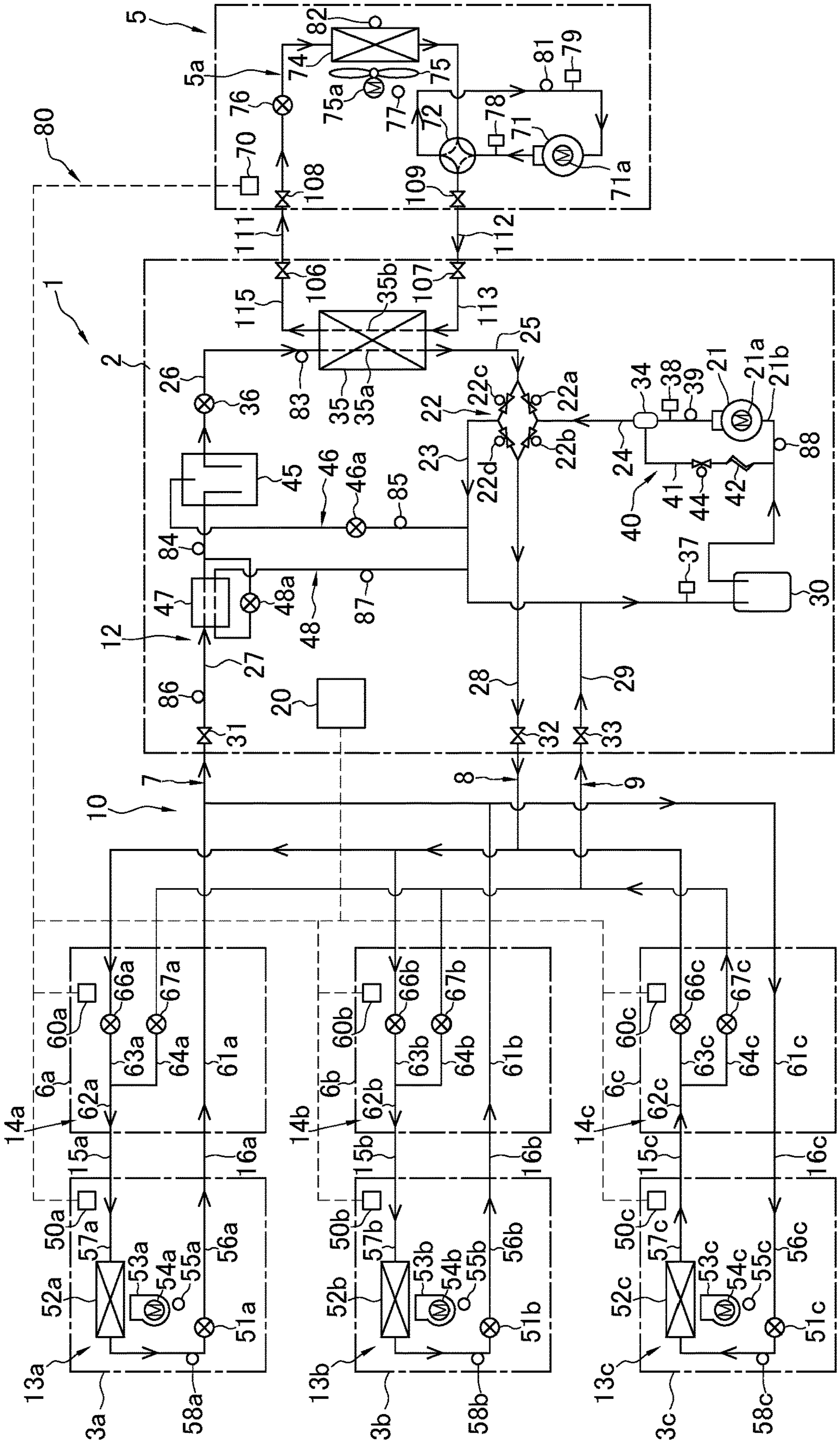

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FIG. 6

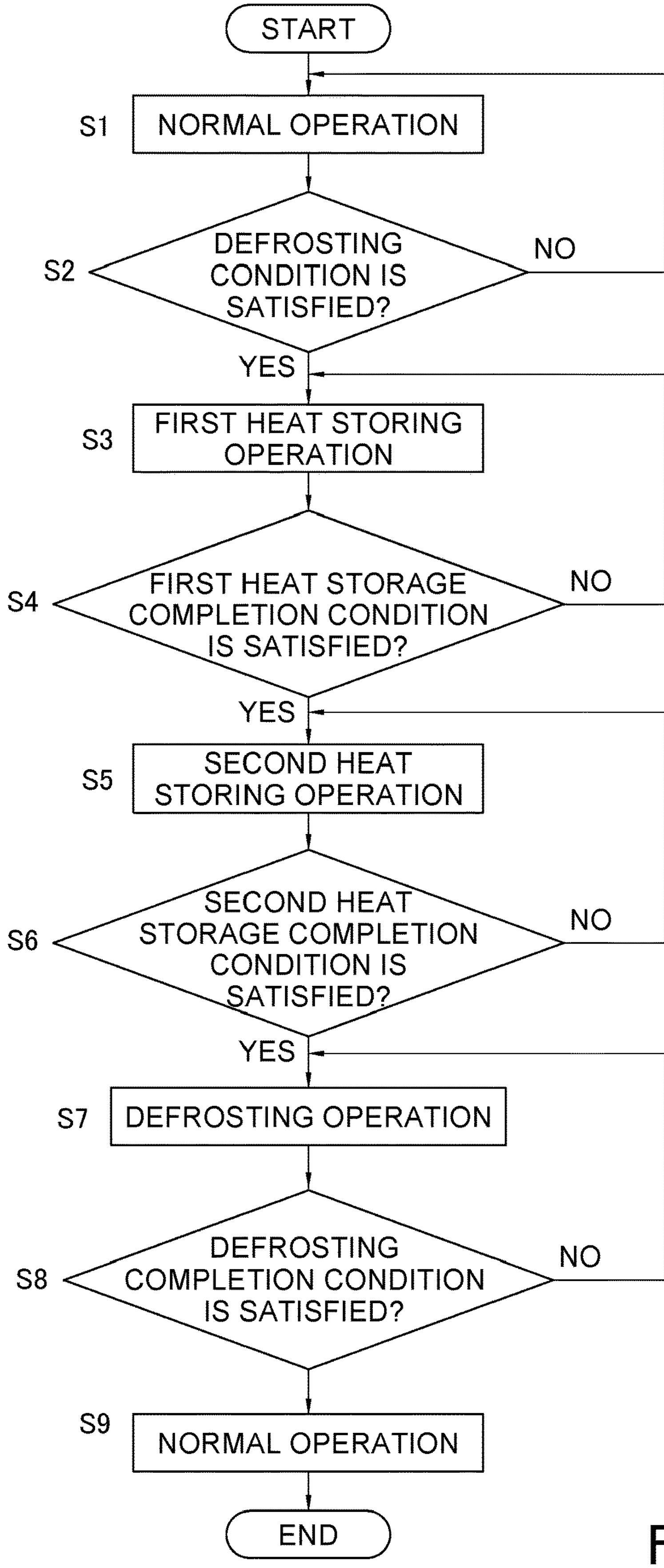
START
S1
NORMAL OPERATION
S2
DEFROSTING CONDITION IS SATISFIED?
NO
YES
S3
FIRST HEAT STORING OPERATION
S4
FIRST HEAT STORAGE COMPLETION CONDITION IS SATISFIED?
NO
YES
S5
SECOND HEAT STORING OPERATION
S6
SECOND HEAT STORAGE COMPLETION CONDITION IS SATISFIED?
NO
YES
S7
DEFROSTING OPERATION
S8
DEFROSTING COMPLETION CONDITION IS SATISFIED?
NO
S9
NORMAL OPERATION
END

FIG. 7

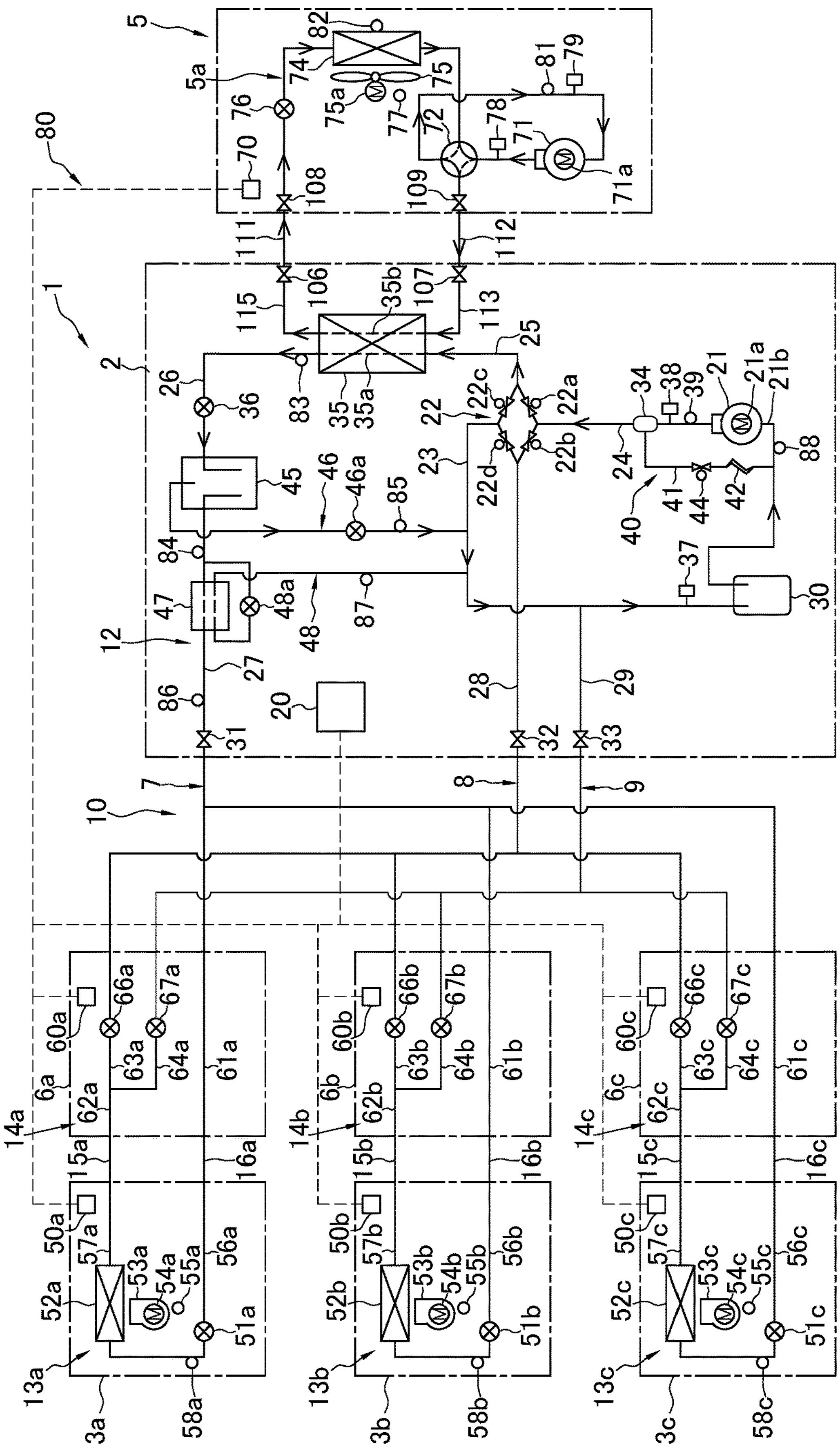

FIG. 8

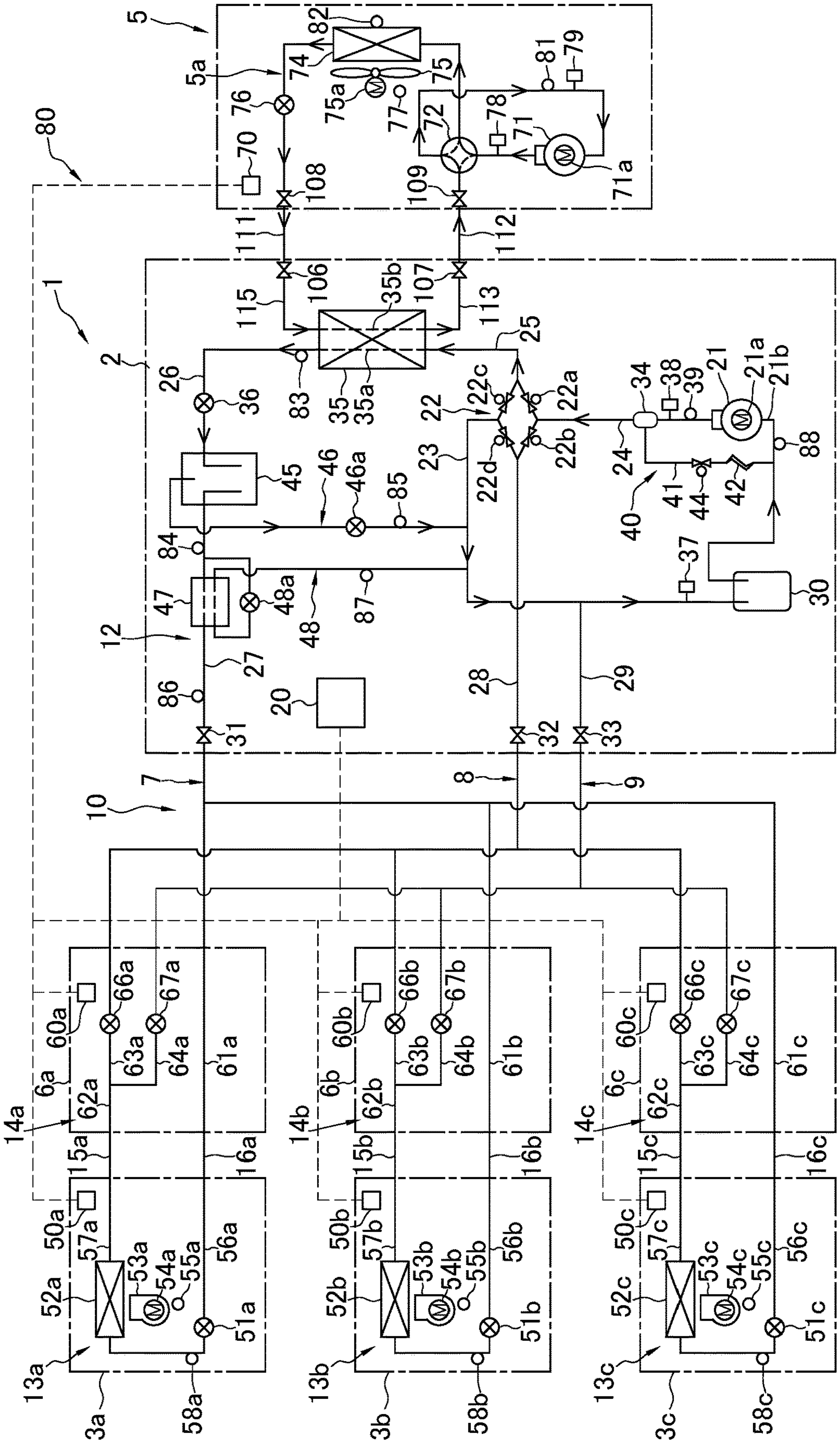

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FIG. 9

REFRIGERATION CYCLE SYSTEM

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of PCT International Application No. PCT/JP2021/043882, filed on Nov. 30, 2021, which claims priority under 35 U.S.C. § 119(a) to Patent Application No. JP 2020-199794, filed in Japan on Dec. 1, 2020, all of which are hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

The present disclosure relates to a refrigeration cycle system.

BACKGROUND ART

There has conventionally been known a binary refrigeration apparatus including a primary refrigerant circuit and a secondary refrigerant circuit connected to each other via a cascade heat exchanger. Such a binary refrigeration apparatus executes defrosting operation in order to melt frost adhering to an evaporator for the primary refrigerant circuit during a heating cycle.

For example, Patent Literature 1 (Japanese Laid-Open Patent Publication No. 2014-109405) discloses a water heating system configured to achieve a heating cycle in a primary refrigerant circuit and a secondary refrigerant circuit to heat water flowing in a water circuit in a secondary heat exchanger. Proposed herein is to cause a refrigerant to flow in a reverse cycle in each of the primary refrigerant circuit and the secondary refrigerant circuit to increase heat quantity supplied to the evaporator of the primary refrigerant circuit during defrosting operation of melting frost adhering to the evaporator of the primary refrigerant circuit.

SUMMARY

A refrigeration cycle system according to a first aspect includes a first circuit and a second circuit. The first circuit allows circulation of the first refrigerant. The first circuit includes a first compressor, a cascade heat exchanger, a heat source heat exchanger, and a first switching unit. The first switching unit is configured to switch a flow path of the first refrigerant. The second circuit allows circulation of a second refrigerant. The second circuit includes a second compressor, the cascade heat exchanger, a utilization heat exchanger, and a second switching unit. The second switching unit is configured to switch a flow path of the second refrigerant. The second circuit includes a bypass circuit and a controlling valve. The bypass circuit connects a portion between the utilization heat exchanger and the cascade heat exchanger, and a suction flow path of the second compressor. The controlling valve is provided on the bypass circuit. The refrigeration cycle system executes defrosting operation. In the defrosting operation, the first refrigerant circulates in the order of the first compressor, the heat source heat exchanger, and the cascade heat exchanger, and the second refrigerant circulates in the order of the second compressor, the cascade heat exchanger, and the bypass circuit.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic configuration diagram of a refrigeration cycle system.

FIG. **2** is a schematic functional block configuration diagram of the refrigeration cycle system.

FIG. **3** is a view indicating behavior (a refrigerant flow) during cooling operation of the refrigeration cycle system.

FIG. **4** is a view indicating behavior (a refrigerant flow) during heating operation of the refrigeration cycle system.

FIG. **5** is a view indicating behavior (a refrigerant flow) during simultaneous cooling and heating operation (mainly cooling) of the refrigeration cycle system.

FIG. **6** is a view indicating behavior (a refrigerant flow) during simultaneous cooling and heating operation (mainly heating) of the refrigeration cycle system.

FIG. **7** is an activation control flowchart of the refrigeration cycle system.

FIG. **8** is a view indicating behavior (a refrigerant flow) during second heat storing operation of the refrigeration cycle system.

FIG. **9** is a view indicating behavior (a refrigerant flow) during defrosting operation of the refrigeration cycle system.

DESCRIPTION OF EMBODIMENTS

(1) Configuration of Refrigeration Cycle System

FIG. **1** is a schematic configuration diagram of a refrigeration cycle system **1**. FIG. **2** is a schematic functional block configuration diagram of the refrigeration cycle system **1**.

The refrigeration cycle system **1** is configured to execute vapor compression refrigeration cycle operation to be used for cooling or heating an indoor space of a building or the like.

The refrigeration cycle system **1** includes a binary refrigerant circuit consisting of a vapor compression primary refrigerant circuit **5***a* (corresponding to a first circuit) and a vapor compression secondary refrigerant circuit **10** (corresponding to a second circuit), and achieves a binary refrigeration cycle. The primary refrigerant circuit **5***a* encloses a refrigerant such as R32 (corresponding to a first refrigerant) for example. The secondary refrigerant circuit **10** encloses a refrigerant such as carbon dioxide (corresponding to a second refrigerant) for example. The primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** are thermally connected via a cascade heat exchanger **35** to be described later.

The refrigeration cycle system **1** includes a primary unit **5**, a heat source unit **2**, a plurality of branching units **6***a*, **6***b*, and **6***c*, and a plurality of utilization units **3***a*, **3***b*, and **3***c*, which are connected correspondingly via pipes. The primary unit **5** and the heat source unit **2** are connected via a primary first connection pipe **111** and a primary second connection pipe **112**. The heat source unit **2** and the plurality of branching units **6***a*, **6***b*, and **6***c* are connected via three refrigerant connection pipes, namely, a secondary second connection pipe **9**, a secondary first connection pipe **8**, and a secondary third connection pipe **7**. The plurality of branching units **6***a*, **6***b*, and **6***c* and the plurality of utilization units **3***a*, **3***b*, and **3***c* are connected via first connecting tubes **15***a*, **15***b*, and **15***c* and second connecting tubes **16***a*, **16***b*, and **16***c*. The present embodiment provides the single primary unit **5**. The present embodiment provides the single heat source unit **2**. The plurality of utilization units **3***a*, **3***b*, and **3***c* according to the present embodiment includes three utilization units, namely, a first utilization unit **3***a*, a second utilization unit **3***b*, and a third utilization unit **3***c*. The plurality of branching units **6***a*, **6***b*, and **6***c* according to the present embodiment includes three branching units, namely, a first branching unit **6***a*, a second branching unit **6***b*, and a third branching unit **6***c*.

In the refrigeration cycle system **1**, the utilization units **3***a*, **3***b*, and **3***c* are configured to individually execute cooling operation or heating operation, and a utilization unit executing heating operation can send a refrigerant to a utilization unit executing cooling operation to achieve heat recovery between the utilization units. Specifically, heat recovery is achieved in the present embodiment by executing mainly cooling operation or mainly heating operation of simultaneously executing cooling operation and heating operation. Furthermore, the refrigeration cycle system **1** is configured to balance a heat load of the heat source unit **2** in accordance with heat loads of all the plurality of utilization units **3***a*, **3***b*, and **3***c* also in consideration of heat recovery mentioned above (mainly cooling operation or mainly heating operation).

(2) Primary Refrigerant Circuit

The primary refrigerant circuit **5***a* includes a primary compressor **71** (corresponding to a first compressor), a primary switching mechanism **72** (corresponding to a first switching unit), a primary heat exchanger **74** (corresponding to a heat source heat exchanger), a primary expansion valve **76**, a first liquid shutoff valve **108**, the primary first connection pipe **111**, a second liquid shutoff valve **106**, a first connecting pipe **115**, the cascade heat exchanger **35** shared with the secondary refrigerant circuit **10**, a second connecting pipe **113**, a second gas shutoff valve **107**, the primary second connection pipe **112**, and a first gas shutoff valve **109**.

The primary compressor **71** is configured to compress a primary refrigerant, and is exemplarily constituted by a positive-displacement compressor of a scroll type or the like configured to inverter control a compressor motor **71***a* to have variable operating capacity.

In a case where the cascade heat exchanger **35** functions as an evaporator for the primary refrigerant, the primary switching mechanism **72** comes into a fifth connection state of connecting a suction side of the primary compressor **71** and a gas side of a primary flow path **35***b* of the cascade heat exchanger **35** (see solid lines in the primary switching mechanism **72** in FIG. **1**). In another case where the cascade heat exchanger **35** functions as a radiator for the primary refrigerant, the primary switching mechanism **72** comes into a sixth connection state of connecting a discharge side of the primary compressor **71** and the gas side of the primary flow path **35***b* of the cascade heat exchanger **35** (see broken lines in the primary switching mechanism **72** in FIG. **1**). The primary switching mechanism **72** is thus configured to switch the flow path of the refrigerant in the primary refrigerant circuit **5***a*, and is exemplarily constituted by a four-way switching valve. With change in switching state of the primary switching mechanism **72**, the cascade heat exchanger **35** can function as the evaporator or the radiator for the primary refrigerant.

The cascade heat exchanger **35** is configured to cause heat exchange between the primary refrigerant such as R32 and a secondary refrigerant such as carbon dioxide without mixing the refrigerants. The cascade heat exchanger **35** is exemplarily constituted by a plate heat exchanger. The cascade heat exchanger **35** includes a secondary flow path **35***a* belonging to the secondary refrigerant circuit **10**, and the primary flow path **35***b* belonging to the primary refrigerant circuit **5***a*. The secondary flow path **35***a* has a gas side connected to a secondary switching mechanism **22** via a third heat source pipe **25**, and a liquid side connected to a heat source expansion valve **36** via a fourth heat source pipe **26**. The primary flow path **35***b* has the gas side connected to the primary compressor **71** via the second connecting pipe **113**, the second gas shutoff valve **107**, the primary second connection pipe **112**, the first gas shutoff valve **109**, and the primary switching mechanism **72**, and a liquid side connected to the second liquid shutoff valve **106** via the first connecting pipe **115**.

The primary heat exchanger **74** is configured to cause heat exchange between the primary refrigerant and outdoor air. The primary heat exchanger **74** has a gas side connected to a pipe extending from the primary switching mechanism **72**. The primary heat exchanger **74** has a liquid side connected to the first liquid shutoff valve **108**. Examples of the primary heat exchanger **74** include a fin-and-tube heat exchanger constituted by large numbers of heat transfer tubes and fins.

The primary expansion valve **76** is provided at a portion between the liquid side of the primary heat exchanger **74** and the first liquid shutoff valve **108**. The primary expansion valve **76** is an electrically powered expansion valve configured to control a flow rate of the primary refrigerant flowing in the primary refrigerant circuit **5***a* and having a controllable opening degree.

The primary first connection pipe **111** is a pipe connecting the first liquid shutoff valve **108** and the second liquid shutoff valve **106**, and connects the primary unit **5** and the heat source unit **2**.

The primary second connection pipe **112** is a pipe connecting the first gas shutoff valve **109** and the second gas shutoff valve **107**, and connects the primary unit **5** and the heat source unit **2**.

The first connecting pipe **115** connects the second liquid shutoff valve **106** and the liquid side of the primary flow path **35***b* of the cascade heat exchanger **35**, and is provided at the heat source unit **2**.

The second connecting pipe **113** connects the gas side of the primary flow path **35***b* of the cascade heat exchanger **35** and the second gas shutoff valve **107**, and is provided at the heat source unit **2**.

The first gas shutoff valve **109** is provided at a portion between the primary second connection pipe **112** and the primary switching mechanism **72**.

(3) Secondary Refrigerant Circuit

The secondary refrigerant circuit **10** includes the plurality of utilization units **3***a*, **3***b*, and **3***c*, the plurality of branching units **6***a*, **6***b*, and **6***c*, and the heat source unit **2**, which are connected correspondingly. Each of the utilization units **3***a*, **3***b*, and **3***c* is connected to a corresponding one of the branching units **6***a*, **6***b*, and **6***c* one by one. Specifically, the utilization unit **3***a* and the branching unit **6***a* are connected via the first connecting tube **15***a* and the second connecting tube **16***a*, the utilization unit **3***b* and the branching unit **6***b* are connected via the first connecting tube **15***b* and the second connecting tube **16***b*, and the utilization unit **3***c* and the branching unit **6***c* are connected via the first connecting tube **15***c* and the second connecting tube **16***c*. Each of the branching units **6***a*, **6***b*, and **6***c* are connected to the heat source unit **2** via three connection pipes, namely, the secondary third connection pipe **7**, the secondary first connection pipe **8**, and the secondary second connection pipe **9**. Specifically, the secondary third connection pipe **7**, the secondary first connection pipe **8**, and the secondary second connection pipe **9** extending from the heat source unit **2** are each branched into a plurality of pipes connected to the branching units **6***a*, **6***b*, and **6***c*.

The secondary first connection pipe **8** has a flow of either the refrigerant in a gas-liquid two-phase state or the refrigerant in a gas state in accordance with an operating state. Depending on the type of the second refrigerant, the secondary first connection pipe **8** has a flow of the refrigerant in a supercritical state in accordance with the operating state. The secondary second connection pipe **9** has a flow of either the refrigerant in the gas-liquid two-phase state or the refrigerant in the gas state in accordance with the operating state. The secondary third connection pipe **7** has a flow of either the refrigerant in the gas-liquid two-phase state or the refrigerant in a liquid state in accordance with the operating state. Depending on the type of the second refrigerant, the secondary third connection pipe **7** has a flow of the refrigerant in the supercritical state in accordance with the operating state.

The secondary refrigerant circuit **10** includes a heat source circuit **12**, branching circuits **14***a*, **14***b*, and **14***c*, and a utilization circuits **13***a*, **13***b*, and **13***c*, which are connected correspondingly.

The heat source circuit **12** principally includes a secondary compressor **21** (corresponding to a second compressor), the secondary switching mechanism **22** (corresponding to a second switching unit), a first heat source pipe **28**, a second heat source pipe **29**, a suction flow path **23**, a discharge flow path **24**, the third heat source pipe **25**, the fourth heat source pipe **26**, a fifth heat source pipe **27**, the cascade heat exchanger **35**, the heat source expansion valve **36**, a third shutoff valve **31**, a first shutoff valve **32**, a second shutoff valve **33**, a secondary accumulator **30**, an oil separator **34**, an oil return circuit **40**, a secondary receiver **45**, a bypass circuit **46** (corresponding to a bypass circuit), a bypass expansion valve **46***a*, a subcooling heat exchanger **47**, a subcooling circuit **48** (corresponding to a bypass circuit), and a subcooling expansion valve **48***a* (corresponding to a controlling valve).

The secondary compressor **21** is configured to compress the secondary refrigerant, and is exemplarily constituted by a positive-displacement compressor of a scroll type or the like configured to inverter control a compressor motor **21***a* to have variable operating capacity. The secondary compressor **21** is controlled in accordance with an operating load so as to have larger operating capacity as the load increases.

The secondary switching mechanism **22** is configured to switch a connection state of the secondary refrigerant circuit **10**, specifically, the flow path of the refrigerant in the heat source circuit **12**. The secondary switching mechanism **22** according to the present embodiment includes four switching valves **22***a*, **22***b*, **22***c*, and **22***d* constituted as two-way valves aligned on an annular flow path. The secondary switching mechanism **22** may alternatively be constituted by a plurality of three-way switching valves combined together. The secondary switching mechanism **22** includes the first switching valve **22***a* provided on a flow path connecting the discharge flow path **24** and the third heat source pipe **25**, the second switching valve **22***b* provided on a flow path connecting the discharge flow path **24** and the first heat source pipe **28**, the third switching valve **22***c* provided on a flow path connecting the suction flow path **23** and the third heat source pipe **25**, and the fourth switching valve **22***d* provided on a flow path connecting the suction flow path **23** and the first heat source pipe **28**. Each of the first switching valve **22***a*, the second switching valve **22***b*, the third switching valve **22***c*, and the fourth switching valve **22***d* according to the present embodiment is an electromagnetic valve configured to be switchable between an opened state and a closed state.

In a case where the cascade heat exchanger **35** functions as a radiator for the secondary refrigerant, the secondary switching mechanism **22** comes into a first connection state of bringing the first switching valve **22***a* into the opened state to connect a discharge side of the secondary compressor **21** and the gas side of the secondary flow path **35***a* of the cascade heat exchanger **35**, and bringing the third switching valve **22***c* into the closed state. In another case where the cascade heat exchanger **35** functions as an evaporator for the secondary refrigerant, the secondary switching mechanism **22** comes into a second connection state of bringing the third switching valve **22***c* into the opened state to connect a suction side of the secondary compressor **21** and the gas side of the secondary flow path **35***a* of the cascade heat exchanger **35**, and bringing the first switching valve **22***a* into the closed state. In a case where the secondary refrigerant discharged from the secondary compressor **21** is sent to the secondary first connection pipe **8**, the secondary switching mechanism **22** comes into a third connection state of bringing the second switching valve **22***b* into the opened state to connect the discharge side of the secondary compressor **21** and the secondary first connection pipe **8**, and bringing the fourth switching valve **22***d* into the closed state. In another case where the refrigerant flowing in the secondary first connection pipe **8** is sucked into the secondary compressor **21**, the secondary switching mechanism **22** comes into a fourth connection state of bringing the fourth switching valve **22***d* into the opened state to connect the secondary first connection pipe **8** and the suction side of the secondary compressor **21**, and bringing the second switching valve **22***b* into the closed state.

As described above, the cascade heat exchanger **35** is configured to cause heat exchange between the primary refrigerant such as R32 and the secondary refrigerant such as carbon dioxide without mixing the refrigerants. The cascade heat exchanger **35** includes the secondary flow path **35***a* having a flow of the secondary refrigerant in the secondary refrigerant circuit **10** and the primary flow path **35***b* having a flow of the primary refrigerant in the primary refrigerant circuit **5***a*, so as to be shared between the primary unit **5** and the heat source unit **2**. The cascade heat exchanger **35** according to the present embodiment is disposed in a heat source casing (not depicted) of the heat source unit **2**. The gas side of the primary flow path **35***b* of the cascade heat exchanger **35** extends to the primary second connection pipe **112** via the second connecting pipe **113** and the second gas shutoff valve **107**. The liquid side of the primary flow path **35***b* of the cascade heat exchanger **35** extends to the primary first connection pipe **111** outside the heat source casing (not depicted) via the first connecting pipe **115** and the second liquid shutoff valve **106**.

The heat source expansion valve **36** is an electrically powered expansion valve having a controllable opening degree and connected to a liquid side of the cascade heat exchanger **35**, in order for control and the like of a flow rate of the secondary refrigerant flowing in the cascade heat exchanger **35**. The heat source expansion valve **36** is provided on the fourth heat source pipe **26**.

Each of the third shutoff valve **31**, the first shutoff valve **32**, and the second shutoff valve **33** is provided at a connecting port with an external device or pipe (specifically, the connection pipe **7**, **8**, or **9**). Specifically, the third shutoff valve **31** is connected to the secondary third connection pipe **7** led out of the heat source unit **2**. The first shutoff valve **32** is connected to the secondary first connection pipe **8** led out of the heat source unit **2**. The second shutoff valve **33** is connected to the secondary second connection pipe **9** led out of the heat source unit **2**.

The first heat source pipe **28** is a refrigerant pipe connecting the first shutoff valve **32** and the secondary switching mechanism **22**. Specifically, the first heat source pipe **28** connects the first shutoff valve **32** and a portion between the second switching valve **22*b*** and the fourth switching valve **22*d*** in the secondary switching mechanism **22**.

The suction flow path **23** connects the secondary switching mechanism **22** and the suction side of the secondary compressor **21**. Specifically, the suction flow path **23** connects a portion between the third switching valve **22*c*** and the fourth switching valve **22*d*** in the secondary switching mechanism **22** and the suction side of the secondary compressor **21**. The suction flow path **23** has a halfway portion provided with the secondary accumulator **30**.

The second heat source pipe **29** is a refrigerant pipe connecting the second shutoff valve **33** and another halfway portion of the suction flow path **23**. The second heat source pipe **29** according to the present embodiment is connected to the suction flow path **23** at a connection point between the portion between the second switching valve **22*b*** and the fourth switching valve **22*d*** in the secondary switching mechanism **22** and the secondary accumulator **30** on the suction flow path **23**.

The discharge flow path **24** is a refrigerant pipe connecting the discharge side of the secondary compressor **21** and the secondary switching mechanism **22**. Specifically, the discharge flow path **24** connects the discharge side of the secondary compressor **21** and a portion between the first switching valve **22*a*** and the second switching valve **22*b*** in the secondary switching mechanism **22**.

The third heat source pipe **25** is a refrigerant pipe connecting the secondary switching mechanism **22** and a gas side of the cascade heat exchanger **35**. Specifically, the third heat source pipe **25** connects a portion between the first switching valve **22*a*** and the third switching valve **22*c*** in the secondary switching mechanism **22** and a gas side end of the secondary flow path **35*a*** in the cascade heat exchanger **35**.

The fourth heat source pipe **26** is a refrigerant pipe connecting the liquid side (opposite to the gas side, and opposite to the side provided with the secondary switching mechanism **22**) of the cascade heat exchanger **35** and the secondary receiver **45**. Specifically, the fourth heat source pipe **26** connects a liquid side end (opposite end to the gas side) of the secondary flow path **35*a*** in the cascade heat exchanger **35** and the secondary receiver **45**.

The secondary receiver **45** is a refrigerant reservoir configured to reserve a residue refrigerant in the secondary refrigerant circuit **10**. The secondary receiver **45** is provided with the fourth heat source pipe **26**, the fifth heat source pipe **27**, and the bypass circuit **46** extending outward.

The bypass circuit **46** is a refrigerant pipe connecting a gas phase region corresponding to an upper region in the secondary receiver **45** and the suction flow path **23**. Specifically, the bypass circuit **46** is connected between the secondary switching mechanism **22** and the secondary accumulator **30** on the suction flow path **23**. The bypass circuit **46** is provided with the bypass expansion valve **46*a***. The bypass expansion valve **46*a*** is an electrically powered expansion valve having a controllable opening degree to control quantity of the refrigerant guided from inside the secondary receiver **45** to the suction side of the secondary compressor **21**.

The fifth heat source pipe **27** is a refrigerant pipe connecting the secondary receiver **45** and the third shutoff valve **31**.

The subcooling circuit **48** is a refrigerant pipe connecting part of the fifth heat source pipe **27** and the suction flow path **23**. Specifically, the subcooling circuit **48** is connected between the secondary switching mechanism **22** and the secondary accumulator **30** on the suction flow path **23**. The subcooling circuit **48** according to the present embodiment extends to branch from a portion between the secondary receiver **45** and the subcooling heat exchanger **47**.

The subcooling heat exchanger **47** is configured to cause heat exchange between the refrigerant flowing in a flow path belonging to the fifth heat source pipe **27** and the refrigerant flowing in a flow path belonging to the subcooling circuit **48**. The subcooling heat exchanger **47** according to the present embodiment is provided between a portion from where the subcooling circuit **48** branches and the third shutoff valve **31** on the fifth heat source pipe **27**. The subcooling expansion valve **48*a*** is provided between a portion branching from the fifth heat source pipe **27** and the subcooling heat exchanger **47** on the subcooling circuit **48**. The subcooling expansion valve **48*a*** is an electrically powered expansion valve having a controllable opening degree and configured to supply the subcooling heat exchanger **47** with a decompressed refrigerant.

The secondary accumulator **30** is a reservoir configured to reserve the secondary refrigerant, and is provided on the suction side of the secondary compressor **21**.

The oil separator **34** is provided at a halfway portion of the discharge flow path **24**. The oil separator **34** is configured to separate refrigerating machine oil discharged from the secondary compressor **21** along with the secondary refrigerant from the secondary refrigerant and return the refrigerating machine oil to the secondary compressor **21**.

The oil return circuit **40** is provided to connect the oil separator **34** and the suction flow path **23**. The oil return circuit **40** includes an oil return flow path **41** as a flow path extending from the oil separator **34** and extending to join a portion between the secondary accumulator **30** and the suction side of the secondary compressor **21** on the suction flow path **23**. The oil return flow path **41** has a halfway portion provided with an oil return capillary tube **42** and an oil return on-off valve **44**. When the oil return on-off valve **44** is controlled into the opened state, the refrigerating machine oil separated in the oil separator **34** passes the oil return capillary tube **42** on the oil return flow path **41** and is returned to the suction side of the secondary compressor **21**. When the secondary compressor **21** is in operation on the secondary refrigerant circuit **10**, the oil return on-off valve **44** according to the present embodiment repetitively is kept in the opened state for predetermined time and is kept in the closed state for predetermined time, to control returned quantity of the refrigerating machine oil through the oil return circuit **40**. The oil return on-off valve **44** according to the present embodiment is an electromagnetic valve controlled to be opened and closed. The oil return on-off valve **44** may alternatively be an electrically powered expansion valve having a controllable opening degree and not provided with the oil return capillary tube **42**. Description is made below to the utilization circuits **13*a***, **13*b***, and **13*c***. As the utilization circuits **13*b*** and **13*c*** are configured similarly to the utilization circuit **13*a***, elements of the utilization circuits **13*b*** and **13*c*** will not be described repeatedly, assuming that a subscript "b" or "c" will replace a subscript "a" in reference signs denoting elements of the utilization circuit **13*a***.

The utilization circuit **13***a* principally includes a utilization heat exchanger **52***a* (corresponding to a utilization heat exchanger), a first utilization pipe **57***a*, a second utilization pipe **56***a*, and a utilization expansion valve **51***a* (corresponding to an expansion valve).

The utilization heat exchanger **52***a* is configured to cause heat exchange between a refrigerant and indoor air, and examples thereof include a fin-and-tube heat exchanger constituted by large numbers of heat transfer tubes and fins. The plurality of utilization heat exchangers **52***a*, **52***b*, and **52***c* are connected in parallel to the secondary switching mechanism **22**, the suction flow path **23**, and the cascade heat exchanger **35**.

The second utilization pipe **56***a* has a first end connected to a liquid side (opposite to a gas side) of the utilization heat exchanger **52***a* in the first utilization unit **3***a*. The second utilization pipe **56***a* has a second end connected to the second connecting tube **16***a*. The second utilization pipe **56***a* has a halfway portion provided with the utilization expansion valve **51***a* described above.

The utilization expansion valve **51***a* is an electrically powered expansion valve configured to control a flow rate of the refrigerant flowing in the utilization heat exchanger **52***a*, and having a controllable opening degree. The utilization expansion valve **51***a* is provided on the second utilization pipe **56***a*.

The first utilization pipe **57***a* has a first end connected to the gas side of the utilization heat exchanger **52***a* in the first utilization unit **3***a*. The first utilization pipe **57***a* according to the present embodiment is connected to a portion opposite to the utilization expansion valve **51***a* of the utilization heat exchanger **52***a*. The first utilization pipe **57***a* has a second end connected to the first connecting tube **15***a*.

Description is made below to the branching circuits **14***a*, **14***b*, and **14***c*. As the branching circuits **14***b* and **14***c* are configured similarly to the branching circuit **14***a*, elements of the branching circuits **14***b* and **14***c* will not be described repeatedly, assuming that a subscript "b" or "c" will replace a subscript "a" in reference signs denoting elements of the branching circuit **14***a*.

The branching circuit **14***a* principally includes a junction pipe **62***a*, a first branching pipe **63***a*, a second branching pipe **64***a*, a first control valve **66***a*, a second control valve **67***a*, and a third branching pipe **61***a*.

The junction pipe **62***a* has a first end connected to the first connecting tube **15***a*. The junction pipe **62***a* has a second end branched to be connected with the first branching pipe **63***a* and the second branching pipe **64***a*.

The first branching pipe **63***a* has a portion opposite to the junction pipe **62***a* and connected to the secondary first connection pipe **8**. The first branching pipe **63***a* is provided with the first control valve **66***a* configured to be opened and closed. The first control valve **66***a* is exemplified herein by an electrically powered expansion valve having a controllable opening degree, but may alternatively be an electromagnetic valve configured only to be opened and closed.

The second branching pipe **64***a* has a portion opposite to the junction pipe **62***a* and connected to the secondary second connection pipe **9**. The second branching pipe **64***a* is provided with the second control valve **67***a* configured to be opened and closed. The second control valve **67***a* is exemplified herein by an electrically powered expansion valve having a controllable opening degree, but may alternatively be an electromagnetic valve configured only to be opened and closed.

The third branching pipe **61***a* has a first end connected to the second connecting tube **16***a*. The third branching pipe **61***a* has a second end connected to the secondary third connection pipe **7**.

During cooling operation to be described later, the first branching unit **6***a* brings the first control valve **66***a* and the second control valve **67***a* into the opened state so as to function as follows. The first branching unit **6***a* sends, to the second connecting tube **16***a*, the refrigerant flowing into the third branching pipe **61***a* via the secondary third connection pipe **7**. The refrigerant flowing in the second utilization pipe **56***a* in the first utilization unit **3***a* via the second connecting tube **16***a* is sent to the utilization heat exchanger **52***a* in the first utilization unit **3***a* via the utilization expansion valve **51***a*. The refrigerant sent to the utilization heat exchanger **52***a* is evaporated by heat exchange with indoor air, and then flows in the first connecting tube **15***a* via the first utilization pipe **57***a*. The refrigerant having flowed in the first connecting tube **15***a* is sent to the junction pipe **62***a* in the first branching unit **6***a*. The refrigerant having flowed in the junction pipe **62***a* is branched into the first branching pipe **63***a* and the second branching pipe **64***a*. The refrigerant having passed the first control valve **66***a* on the first branching pipe **63***a* is sent to the secondary first connection pipe **8**. The refrigerant having passed the second control valve **67***a* on the second branching pipe **64***a* is sent to the secondary second connection pipe **9**.

In a case where the first utilization unit **3***a* cools the indoor space during mainly cooling operation and mainly heating operation to be described later, the first branching unit **6***a* brings the first control valve **66***a* into the closed state and the second control valve **67***a* into the opened state so as to function as follows. The first branching unit **6***a* sends, to the second connecting tube **16***a*, the refrigerant flowing into the third branching pipe **61***a* via the secondary third connection pipe **7**. The refrigerant flowing in the second utilization pipe **56***a* in the first utilization unit **3***a* via the second connecting tube **16***a* is sent to the utilization heat exchanger **52***a* in the first utilization unit **3***a* via the utilization expansion valve **51***a*. The refrigerant sent to the utilization heat exchanger **52***a* is evaporated by heat exchange with indoor air, and then flows in the first connecting tube **15***a* via the first utilization pipe **57***a*. The refrigerant having flowed in the first connecting tube **15***a* is sent to the junction pipe **62***a* in the first branching unit **6***a*. The refrigerant having flowed in the junction pipe **62***a* flows to the second branching pipe **64***a* and passes the second control valve **67***a* to be subsequently sent to the secondary second connection pipe **9**.

During heating operation to be described later, the first branching unit **6***a* brings the second control valve **67***a* into the opened or closed state and brings the first control valve **66***a* into the opened state in accordance with an operation condition so as to function as follows. In the first branching unit **6***a*, the refrigerant flowing into the first branching pipe **63***a* via the secondary first connection pipe **8** passes the first control valve **66***a* to be sent to the junction pipe **62***a*. The refrigerant having flowed in the junction pipe **62***a* flows in the first utilization pipe **57***a* in the utilization unit **3***a* via the first connecting tube **15***a* to be sent to the utilization heat exchanger **52***a*. The refrigerant sent to the utilization heat exchanger **52***a* radiates heat through heat exchange with indoor air, and then passes the utilization expansion valve **51***a* provided on the second utilization pipe **56***a*. The refrigerant having passed the second utilization pipe **56***a* flows in the third branching pipe **61***a* in the first branching unit **6***a* via the second connecting tube **16***a* to be subsequently sent to the secondary third connection pipe **7**.

In another case where the first utilization unit **3***a* heats the indoor space during mainly cooling operation and mainly heating operation to be described later, the first branching unit **6***a* brings the second control valve **67***a* into the closed state and brings the first control valve **66***a* into the opened state so as to function as follows. In the first branching unit **6***a*, the refrigerant flowing into the first branching pipe **63***a* via the secondary first connection pipe **8** passes the first control valve **66***a* to be sent to the junction pipe **62***a*. The refrigerant having flowed in the junction pipe **62***a* flows in the first utilization pipe **57***a* in the utilization unit **3***a* via the first connecting tube **15***a* to be sent to the utilization heat exchanger **52***a*. The refrigerant sent to the utilization heat exchanger **52***a* radiates heat through heat exchange with indoor air, and then passes the utilization expansion valve **51***a* provided on the second utilization pipe **56***a*. The refrigerant having passed the second utilization pipe **56***a* flows in the third branching pipe **61***a* in the first branching unit **6***a* via the second connecting tube **16***a* to be subsequently sent to the secondary third connection pipe **7**.

The first branching unit **6***a*, as well as the second branching unit **6***b* and the third branching unit **6***c*, similarly have such a function. Accordingly, the first branching unit **6***a*, the second branching unit **6***b*, and the third branching unit **6***c* are configured to individually switchably cause the utilization heat exchangers **52***a*, **52***b*, and **52***c* to function as a refrigerant evaporator or a refrigerant radiator.

(4) Primary Unit

The primary unit **5** is disposed in a space different from a space provided with the utilization units **3***a*, **3***b*, and **3***c* and the branching units **6***a*, **6***b*, and **6***c*, on a roof, or the like.

The primary unit **5** includes part of the primary refrigerant circuit **5***a* described above, a primary fan **75**, various sensors, and a primary control unit **70**, which are accommodated in a primary casing (not depicted).

The primary unit **5** includes, as the part of the primary refrigerant circuit **5***a*, the primary compressor **71**, the primary switching mechanism **72**, the primary heat exchanger **74**, the primary expansion valve **76**, the first liquid shutoff valve **108**, and the first gas shutoff valve **109**.

The primary fan **75** is provided in the primary unit **5**, and is configured to generate an air flow of guiding outdoor air into the primary heat exchanger **74**, and exhausting, to outdoors, air obtained after heat exchange with the primary refrigerant flowing in the primary heat exchanger **74**. The primary fan **75** is driven by a primary fan motor **75***a*.

The primary unit **5** is provided with the various sensors. Specifically, there are provided an outdoor air temperature sensor **77** configured to detect temperature of outdoor air to be subject to pass the primary heat exchanger **74**, a primary discharge pressure sensor **78** configured to detect pressure of the primary refrigerant discharged from the primary compressor **71**, a primary suction pressure sensor **79** configured to detect pressure of the primary refrigerant sucked into the primary compressor **71**, a primary suction temperature sensor **81** configured to detect temperature of the primary refrigerant sucked into the primary compressor **71**, and a primary heat-exchange temperature sensor **82** configured to detect temperature of the refrigerant flowing in the primary heat exchanger **74**.

The primary control unit **70** controls behavior of the elements **71** (**71***a*), **72**, **75** (**75***a*), and **76** provided in the primary unit **5**. The primary control unit **70** includes a processor such as a CPU or a microcomputer or the like provided to control the primary unit **5** and a memory, so as to transmit and receive control signals and the like to and from a remote controller (not depicted), and to transmit and receive control signals and the like among a heat source control unit **20**, branching unit control units **60***a*, **60***b*, and **60***c*, and utilization control units **50***a*, **50***b*, and **50***c*.

(5) Heat Source Unit

The heat source unit **2** is disposed in a space different from the space provided with the utilization units **3***a*, **3***b*, and **3***c* and the branching units **6***a*, **6***b*, and **6***c*, on a roof, or the like.

The heat source unit **2** is connected to the branching units **6***a*, **6***b*, and **6***c* via the connection pipes **7**, **8**, and **9**, to constitute part of the secondary refrigerant circuit **10**. The heat source unit **2** is connected with the primary unit **5** via the primary first connection pipe **111** and the primary second connection pipe **112**, to constitute part of the primary refrigerant circuit **5***a*.

The heat source unit **2** principally includes the heat source circuit **12** described above, various sensors, the heat source control unit **20**, the second liquid shutoff valve **106** constituting part of the primary refrigerant circuit **5***a*, the first connecting pipe **115**, the second connecting pipe **113**, and the second gas shutoff valve **107**, which are accommodated in the heat source casing (not depicted).

The heat source unit **2** is provided with a secondary suction pressure sensor **37** configured to detect pressure of the secondary refrigerant on the suction side of the secondary compressor **21**, a secondary discharge pressure sensor **38** configured to detect pressure of the secondary refrigerant on the discharge side of the secondary compressor **21**, a secondary discharge temperature sensor **39** configured to detect temperature of the secondary refrigerant on the discharge side of the secondary compressor **21**, a secondary suction temperature sensor **88** configured to detect temperature of the secondary refrigerant on the suction side of the secondary compressor **21**, a secondary cascade temperature sensor **83** configured to detect temperature of the secondary refrigerant flowing between the secondary flow path **35***a* of the cascade heat exchanger **35** and the heat source expansion valve **36**, a receiver outlet temperature sensor **84** configured to detect temperature of the secondary refrigerant flowing between the secondary receiver **45** and the subcooling heat exchanger **47**, a bypass circuit temperature sensor **85** configured to detect temperature of the secondary refrigerant flowing downstream of the bypass expansion valve **46***a* on the bypass circuit **46**, a subcooling outlet temperature sensor **86** configured to detect temperature of the secondary refrigerant flowing between the subcooling heat exchanger **47** and the third shutoff valve **31**, and a subcooling circuit temperature sensor **87** configured to detect temperature of the secondary refrigerant flowing at an outlet of the subcooling heat exchanger **47** on the subcooling circuit **48**.

The heat source control unit **20** controls behavior of the elements **21** (**21***a*), **22**, **36**, **44**, **46***a*, and **48***a* provided in the heat source unit **2**. The heat source control unit **20** includes a processor such as a CPU or a microcomputer or the like provided to control the heat source unit **2** and a memory, so as to transmit and receive control signals and the like among the primary control unit **70** in the primary unit **5**, the utilization control units **50***a*, **50***b*, and **50***c* in the utilization units **3***a*, **3***b*, and **3***c*, and the branching unit control units **60***a*, **60***b*, and **60***c*.

(6) Utilization Unit

The utilization units **3***a*, **3***b*, and **3***c* are installed by being embedded in or being suspended from a ceiling in an indoor space of a building or the like, or by being hung on a wall surface in the indoor space, or the like.

The utilization units **3**a, **3**b, and **3**c are connected to the heat source unit **2** via the connection pipes **7**, **8**, and **9**.

The utilization units **3**a, **3**b, and **3**c respectively include the utilization circuits **13**a, **13**b, and **13**c constituting part of the secondary refrigerant circuit **10**.

The utilization units **3**a, **3**b, and **3**c will be described hereinafter in terms of their configurations. The second utilization unit **3**b and the third utilization unit **3**c are configured similarly to the first utilization unit **3**a. The configuration of only the first utilization unit **3**a will thus be described herein. As to the configuration of each of the second utilization unit **3**b and the third utilization unit **3**c, elements will be denoted by reference signs obtained by replacing a subscript "a" in reference signs of elements of the first utilization unit **3**a with a subscript "b" or "c", and these elements will not be described repeatedly.

The first utilization unit **3**a principally includes the utilization circuit **13**a described above, an indoor fan **53**a, the utilization control unit **50**a, and various sensors. The indoor fan **53**a includes an indoor fan motor **54**a.

The indoor fan **53**a generates an air flow of sucking indoor air into the unit and supplying the indoor space with supply air obtained after heat exchange with the refrigerant flowing in the utilization heat exchanger **52**a. The indoor fan **53**a is driven by the indoor fan motor **54**a.

The utilization unit **3**a is provided with a liquid-side temperature sensor **58**a configured to detect temperature of a refrigerant on the liquid side of the utilization heat exchanger **52**a. The utilization unit **3**a is further provided with an indoor temperature sensor **55**a configured to detect indoor temperature as temperature of air introduced from the indoor space and to be subject to pass the utilization heat exchanger **52**a.

The utilization control unit **50**a controls behavior of the elements **51**a and **53**a (**54**a) of the utilization unit **3**a. The utilization control unit **50**a includes a processor such as a CPU or a microcomputer or the like provided to control the utilization unit **3**a and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not depicted), and to transmit and receive control signals and the like among the heat source control unit **20**, the branching unit control units **60**a, **60**b, and **60**c, and the primary control unit **70** in the primary unit **5**.

The second utilization unit **3**b includes the utilization circuit **13**b, an indoor fan **53**b, the utilization control unit **50**b, and an indoor fan motor **54**b. The third utilization unit **3**c includes the utilization circuit **13**c, an indoor fan **53**c, the utilization control unit **50**c, and an indoor fan motor **54**c.

(7) Branching Unit

The branching units **6**a, **6**b, and **6**c are installed in a space behind the ceiling of the indoor space of the building or the like.

Each of the branching units **6**a, **6**b, and **6**c is connected to a corresponding one of the utilization units **3**a, **3**b, and **3**c one by one. The branching units **6**a, **6**b, and **6**c are connected to the heat source unit **2** via the connection pipes **7**, **8**, and **9**.

The branching units **6**a, **6**b, and **6**c will be described next in terms of their configurations. The second branching unit **6**b and the third branching unit **6**c are configured similarly to the first branching unit **6**a. The configuration of only the first branching unit **6**a will thus be described herein. As to the configuration of each of the second branching unit **6**b and the third branching unit **6**c, elements will be denoted by reference signs obtained by replacing a subscript "a" in reference signs of elements of the first branching unit **6**a with a subscript "b" or "c", and these elements will not be described repeatedly.

The first branching unit **6**a principally includes the branching circuit **14**a described above, and the branching unit control unit **60**a.

The branching unit control unit **60**a controls behavior of the elements **66**a and **67**a of the branching unit **6**a. The branching unit control unit **60**a includes a processor such as a CPU or a microcomputer or the like provided to control the branching unit **6**a and a memory, so as to transmit and receive control signals and the like to and from the remote controller (not depicted), and to transmit and receive control signals and the like among the heat source control unit **20**, the utilization units **3**a, **3**b, and **3**c, and the primary control unit **70** in the primary unit **5**.

The second branching unit **6**b includes the branching circuit **14**b, and the branching unit control unit **60**b. The third branching unit **6**c includes the branching circuit **14**c, and the branching unit control unit **60**c.

(8) Control Unit

In the refrigeration cycle system **1**, the heat source control unit **20**, the utilization control units **50**a, **50**b, and **50**c, the branching unit control units **60**a, **60**b, and **60**c, and the primary control unit **70** described above are connected wiredly or wirelessly to be mutually communicable so as to constitute a control unit **80**. The control unit **80** accordingly controls behavior of the elements **21** (**21**a), **22**, **36**, **44**, **46**a, **48**a, **51**a, **51**b, **51**c, **53**a, **53**b, **53**c (**54**a, **54**b, **54**c), **66**a, **66**b, **66**c, **67**a, **67**b, **67**c, **71** (**71**a), **72**, **75** (**75**a), and **76** in accordance with detection information of the various sensors **37**, **38**, **39**, **83**, **84**, **85**, **86**, **87**, **88**, **77**, **78**, **79**, **81**, **82**, **58**a, **58**b, **58**c, and the like, command information received from the remote controller (not depicted), and the like.

(9) Behavior of Refrigeration Cycle System

Behavior of the refrigeration cycle system **1** will be described next with reference to FIG. **3** to FIG. **6**.

Refrigeration cycle operation of the refrigeration cycle system **1** can be divided principally into cooling operation, heating operation, mainly cooling operation, and mainly heating operation. During heating operation and mainly heating operation, heat storing operation and defrosting operation to be described later are executed if a predetermined condition is satisfied.

Herein, cooling operation corresponds to refrigeration cycle operation in a case where there are only utilization units each of which operates with the utilization heat exchanger functioning as a refrigerant evaporator, and the cascade heat exchanger **35** functions as a radiator for the secondary refrigerant with respect to evaporation loads of all the utilization units.

Heating operation corresponds to refrigeration cycle operation in a case where there are only utilization units each of which operates with the utilization heat exchanger functioning as a refrigerant radiator, and the cascade heat exchanger **35** functions as an evaporator for the secondary refrigerant with respect to radiation loads of all the utilization units.

During mainly cooling operation, there coexist a utilization unit operating with the utilization heat exchanger functioning as a refrigerant evaporator and a utilization unit operating with the utilization heat exchanger functioning as a refrigerant radiator. Mainly cooling operation corresponds to refrigeration cycle operation in a case where the cascade heat exchanger **35** functions as a radiator for the secondary refrigerant with respect to evaporation loads of all the utilization units principally occupying heat loads of all the utilization units.

During mainly heating operation, there coexist a utilization unit operating with the utilization heat exchanger functioning as a refrigerant evaporator, and a utilization unit operating with the utilization heat exchanger functioning as a refrigerant radiator. Mainly heating operation corresponds to refrigeration cycle operation in a case where the cascade heat exchanger **35** functions as an evaporator for the secondary refrigerant with respect to radiation loads of all the utilization units principally occupying heat loads of all the utilization units.

Behavior of the refrigeration cycle system **1** including these types of refrigeration cycle operation is executed by the control unit **80**.

(9-1) Cooling Operation

During cooling operation, for example, each of the utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* functions as a refrigerant evaporator, and the cascade heat exchanger **35** functions as a radiator for the secondary refrigerant. During such cooling operation, the primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** in the refrigeration cycle system **1** are configured as depicted in FIG. **3**. FIG. **3** includes arrows provided to the primary refrigerant circuit **5***a* and arrows provided to the secondary refrigerant circuit **10**, which indicate refrigerant flows during cooling operation.

Specifically, in the primary unit **5**, the primary switching mechanism **72** is switched into the fifth connection state to cause the cascade heat exchanger **35** to function as an evaporator for the primary refrigerant. The fifth connection state of the primary switching mechanism **72** is depicted by solid lines in the primary switching mechanism **72** in FIG. **3**. Accordingly in the primary unit **5**, the primary refrigerant discharged from the primary compressor **71** passes the primary switching mechanism **72** and exchanges heat with outdoor air supplied from the primary fan **75** in the primary heat exchanger **74** to be condensed. The primary refrigerant condensed in the primary heat exchanger **74** is decompressed at the primary expansion valve **76**, then flows in the primary flow path **35***b* of the cascade heat exchanger **35** to be evaporated, and is sucked into the primary compressor **71** via the primary switching mechanism **72**.

In the heat source unit **2**, the secondary switching mechanism **22** is switched into the first connection state as well as the fourth connection state to cause the cascade heat exchanger **35** to function as a radiator for the secondary refrigerant. In the first connection state of the secondary switching mechanism **22**, the first switching valve **22***a* is in the opened state and the third switching valve **22***c* is in the closed state. In the fourth connection state of the secondary switching mechanism **22**, the fourth switching valve **22***d* is in the opened state and the second switching valve **22***b* is in the closed state. The heat source expansion valve **36** is controlled in opening degree. In the first to third utilization units **3***a*, **3***b*, and **3***c*, the first control valves **66***a*, **66***b*, and **66***c* and the second control valves **67***a*, **67***b*, and **67***c* are controlled into the opened state. Accordingly, each of the utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* functions as a refrigerant evaporator. All the utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* and the suction side of the secondary compressor **21** in the heat source unit **2** are connected via the first utilization pipes **57***a*, **57***b*, and **57***c*, the first connecting tubes **15***a*, **15***b*, and **15***c*, the junction pipes **62***a*, **62***b*, and **62***c*, the first branching pipes **63***a*, **63***b*, and **63***c*, the second branching pipes **64***a*, **64***b*, and **64***c*, the first connection pipe **8**, and the second connection pipe **9**. The subcooling expansion valve **48***a* is controlled in opening degree such that the secondary refrigerant flowing at the outlet of the subcooling heat exchanger **47** toward the third connection pipe **7** has a degree of subcooling at a predetermined value. The bypass expansion valve **46***a* is controlled into the closed state. In the utilization units **3***a*, **3***b*, and **3***c*, the utilization expansion valves **51***a*, **51***b*, and **51***c* are each controlled in opening degree.

In the secondary refrigerant circuit **10** in this state, a secondary high-pressure refrigerant compressed in and discharged from the secondary compressor **21** is sent to the secondary flow path **35***a* of the cascade heat exchanger **35** via the secondary switching mechanism **22**. The secondary high-pressure refrigerant flowing in the secondary flow path **35***a* of the cascade heat exchanger **35** radiates heat, and the primary refrigerant flowing in the primary flow path **35***b* of the cascade heat exchanger **35** is evaporated. The secondary refrigerant having radiated heat in the cascade heat exchanger **35** passes the heat source expansion valve **36** controlled in opening degree, and then flows into the receiver **45**. Part of the refrigerant having flowed out of the receiver **45** is branched into the subcooling circuit **48**, is decompressed at the subcooling expansion valve **48***a*, and then joins the suction flow path **23**. In the subcooling heat exchanger **47**, part of the remaining refrigerant having flowed out of the receiver **45** is cooled by the refrigerant flowing in the subcooling circuit **48**, and is then sent to the third connection pipe **7** via the third shutoff valve **31**.

The refrigerant sent to the third connection pipe **7** is branched into three portions to pass the third branching pipes **61***a*, **61***b*, and **61***c* of the first to third branching units **6***a*, **6***b*, and **6***c*. Thereafter, the refrigerant having flowed in the second connecting tubes **16***a*, **16***b*, and **16***c* is sent to the second utilization pipes **56***a*, **56***b*, and **56***c* of the first to third utilization units **3***a*, **3***b*, and **3***c*. The refrigerant sent to the second utilization pipes **56***a*, **56***b*, and **56***c* is sent to the utilization expansion valves **51***a*, **51***b*, and **51***c* in the utilization units **3***a*, **3***b*, and **3***c*.

The refrigerant having passed the utilization expansion valves **51***a*, **51***b*, and **51***c* each controlled in opening degree exchanges heat with indoor air supplied by the indoor fans **53***a*, **53***b*, and **53***c* in the utilization heat exchangers **52***a*, **52***b*, and **52***c*. The refrigerant flowing in the utilization heat exchangers **52***a*, **52***b*, and **52***c* is thus evaporated into a low-pressure gas refrigerant. Indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization heat exchangers **52***a*, **52***b*, and **52***c* flows in the first utilization pipes **57***a*, **57***b*, and **57***c*, flows in the first connecting tubes **15***a*, **15***b*, and **15***c*, and is then sent to the junction pipes **62***a*, **62***b*, and **62***c* of the first to third branching units **6***a*, **6***b*, and **6***c*.

The low-pressure gas refrigerant sent to the junction pipes **62***a*, **62***b*, and **62***c* is branched into the first branching pipes **63***a*, **63***b*, and **63***c*, and the second branching pipes **64***a*, **64***b*, and **64***c* and flows. The refrigerant having passed the first control valves **66***a*, **66***b*, and **66***c* on the first branching pipes **63***a*, **63***b*, and **63***c* is sent to the first connection pipe **8**. The refrigerant having passed the second control valves **67***a*,

67*b*, and **67***c* on the second branching pipes **64***a*, **64***b*, and **64***c* is sent to the second connection pipe **9**.

The low-pressure gas refrigerant sent to the first connection pipe **8** and the second connection pipe **9** is returned to the suction side of the secondary compressor **21** via the first shutoff valve **32**, the second shutoff valve **33**, the first heat source pipe **28**, the second heat source pipe **29**, the secondary switching mechanism **22**, the suction flow path **23**, and the accumulator **30**.

Behavior during cooling operation is executed in this manner.

(9-2) Heating Operation

During heating operation, each of the utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* functions as a refrigerant radiator. Furthermore, during heating operation, the cascade heat exchanger **35** functions as an evaporator for the secondary refrigerant. During heating operation, the primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** in the refrigeration cycle system **1** are configured as depicted in FIG. **4**. FIG. **4** includes arrows provided to the primary refrigerant circuit **5***a* and arrows provided to the secondary refrigerant circuit **10**, which indicate refrigerant flows during heating operation.

Specifically, in the primary unit **5**, the primary switching mechanism **72** is switched into a sixth connection state to cause the cascade heat exchanger **35** to function as a radiator for the primary refrigerant. The sixth connection state of the primary switching mechanism **72** corresponds to a connection state depicted by broken lines in the primary switching mechanism **72** in FIG. **4**. Accordingly in the primary unit **5**, the primary refrigerant discharged from the primary compressor **71** passes the primary switching mechanism **72** and flows in the primary flow path **35***b* of the cascade heat exchanger **35** to be condensed. The primary refrigerant condensed in the cascade heat exchanger **35** is decompressed at the primary expansion valve **76**, then exchanges heat with outdoor air supplied from the primary fan **75** in the primary heat exchanger **74** to be evaporated, and is sucked into the primary compressor **71** via the primary switching mechanism **72**.

In the heat source unit **2**, the secondary switching mechanism **22** is switched into the second connection state as well as the third connection state. The cascade heat exchanger **35** thus functions as an evaporator for the secondary refrigerant. In the second connection state of the secondary switching mechanism **22**, the first switching valve **22***a* is in the closed state and the third switching valve **22***c* is in the opened state. In the third connection state of the secondary switching mechanism **22**, the second switching valve **22***b* is in the opened state and the fourth switching valve **22***d* is in the closed state. The heat source expansion valve **36** is controlled in opening degree. In the first to third branching units **6***a*, **6***b*, and **6***c*, the first control valves **66***a*, **66***b*, and **66***c* are controlled into the opened state, and the second control valves **67***a*, **67***b*, and **67***c* are controlled into the closed state. Accordingly, each of the utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* functions as a refrigerant radiator. The utilization heat exchangers **52***a*, **52***b*, and **52***c* in the utilization units **3***a*, **3***b*, and **3***c* and the discharge side of the secondary compressor **21** in the heat source unit **2** are connected via the discharge flow path **24**, the first heat source pipe **28**, the first connection pipe **8**, the first branching pipes **63***a*, **63***b*, and **63***c*, the junction pipes **62***a*, **62***b*, and **62***c*, the first connecting tubes **15***a*, **15***b*, and **15***c*, and the first utilization pipes **57***a*, **57***b*, and **57***c*. The subcooling expansion valve **48***a* and the bypass expansion valve **46***a* are controlled into the closed state. In the utilization units **3***a*, **3***b*, and **3***c*, the utilization expansion valves **51***a*, **51***b*, and **51***c* are each controlled in opening degree.

In the secondary refrigerant circuit **10** in this state, a high-pressure refrigerant compressed in and discharged from the secondary compressor **21** is sent to the first heat source pipe **28** via the second switching valve **22***b* controlled into the opened state in the secondary switching mechanism **22**. The refrigerant sent to the first heat source pipe **28** is sent to the first connection pipe **8** via the first shutoff valve **32**.

The high-pressure refrigerant sent to the first connection pipe **8** is branched into three portions to be sent to the first branching pipes **63***a*, **63***b*, and **63***c* in the utilization units **3***a*, **3***b*, and **3***c* in operation. The high-pressure refrigerant sent to the first branching pipes **63***a*, **63***b*, and **63***c* passes the first control valves **66***a*, **66***b*, and **66***c*, and flows in the junction pipes **62***a*, **62***b*, and **62***c*. The refrigerant having flowed in the first connecting tubes **15***a*, **15***b*, and **15***c* and the first utilization pipes **57***a*, **57***b*, and **57***c* is then sent to the utilization heat exchangers **52***a*, **52***b*, and **52***c*.

The high-pressure refrigerant sent to the utilization heat exchangers **52***a*, **52***b*, and **52***c* exchanges heat with indoor air supplied by the indoor fans **53***a*, **53***b*, and **53***c* in the utilization heat exchangers **52***a*, **52***b*, and **52***c*. The refrigerant flowing in the utilization heat exchangers **52***a*, **52***b*, and **52***c* thus radiates heat. Indoor air is heated and is supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization heat exchangers **52***a*, **52***b*, and **52***c* flows in the second utilization pipes **56***a*, **56***b*, and **56***c* and passes the utilization expansion valves **51***a*, **51***b*, and **51***c* each controlled in opening degree. Thereafter, the refrigerant having flowed in the second connecting tubes **16***a*, **16***b*, and **16***c* flows in the third branching pipes **61***a*, **61***b*, and **61***c* of the branching units **6***a*, **6***b*, and **6***c*.

The refrigerant sent to the third branching pipes **61***a*, **61***b*, and **61***c* is sent to the third connection pipe **7** to join.

The refrigerant sent to the third connection pipe **7** is sent to the heat source expansion valve **36** via the third shutoff valve **31**. The refrigerant sent to the heat source expansion valve **36** is controlled in flow rate at the heat source expansion valve **36** and is then sent to the cascade heat exchanger **35**. In the cascade heat exchanger **35**, the secondary refrigerant flowing in the secondary flow path **35***a* is evaporated into a low-pressure gas refrigerant and is sent to the secondary switching mechanism **22**, and the primary refrigerant flowing in the primary flow path **35***b* of the cascade heat exchanger **35** is condensed. The secondary low-pressure gas refrigerant sent to the secondary switching mechanism **22** is returned to the suction side of the secondary compressor **21** via the suction flow path **23** and the accumulator **30**.

Behavior during heating operation is executed in this manner.

(9-3) Mainly Cooling Operation

During mainly cooling operation, for example, the utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* each function as a refrigerant evaporator, and the utilization heat exchanger **52***c* in the utilization unit **3***c* functions as a refrigerant radiator. During mainly cooling operation, the cascade heat exchanger **35** functions as a radiator for the secondary refrigerant. During mainly cooling operation, the primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** in the refrigeration cycle system **1** are configured as depicted in FIG. **5**. FIG. **5** includes arrows provided to the primary refrigerant circuit **5***a* and arrows provided to the secondary refrigerant circuit **10**, which indicate refrigerant flows during mainly cooling operation.

Specifically, in the primary unit **5**, the primary switching mechanism **72** is switched into the fifth connection state (the state depicted by solid lines in the primary switching mechanism **72** in FIG. **5**) to cause the cascade heat exchanger **35** to function as an evaporator for the primary refrigerant. Accordingly in the primary unit **5**, the primary refrigerant discharged from the primary compressor **71** passes the primary switching mechanism **72** and exchanges heat with outdoor air supplied from the primary fan **75** in the primary heat exchanger **74** to be condensed. The primary refrigerant condensed in the primary heat exchanger **74** is decompressed at the primary expansion valve **76**, then flows in the primary flow path **35***b* of the cascade heat exchanger **35** to be evaporated, and is sucked into the primary compressor **71** via the primary switching mechanism **72**.

In the heat source unit **2**, the secondary switching mechanism **22** is switched into the first connection state (the first switching valve **22***a* is in the opened state and the third switching valve **22***c* is in the closed state) as well as the third connection state (the second switching valve **22***b* is in the opened state and the fourth switching valve **22***d* is in the closed state) to cause the cascade heat exchanger **35** to function as a radiator for the secondary refrigerant. The heat source expansion valve **36** is controlled in opening degree. In the first to third branching units **6***a*, **6***b*, and **6***c*, the first control valve **66***c* and the second control valves **67***a* and **67***b* are controlled into the opened state, and the first control valves **66***a* and **66***b* and the second control valve **67***c* are controlled into the closed state. Accordingly, the utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* each function as a refrigerant evaporator, and the utilization heat exchanger **52***c* in the utilization unit **3***c* functions as a refrigerant radiator. The utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* and the suction side of the secondary compressor **21** in the heat source unit **2** are connected via the second connection pipe **9**, and the utilization heat exchanger **52***c* in the utilization unit **3***c* and the discharge side of the secondary compressor **21** in the heat source unit **2** are connected via the first connection pipe **8**. The subcooling expansion valve **48***a* is controlled in opening degree such that the secondary refrigerant flowing at the outlet of the subcooling heat exchanger **47** toward the third connection pipe **7** has a degree of subcooling at a predetermined value. The bypass expansion valve **46***a* is controlled into the closed state. In the utilization units **3***a*, **3***b*, and **3***c*, the utilization expansion valves **51***a*, **51***b*, and **51***c* are each controlled in opening degree.

In the secondary refrigerant circuit **10** in this state, part of the secondary high-pressure refrigerant compressed in and discharged from the secondary compressor **21** is sent to the first connection pipe **8** via the secondary switching mechanism **22**, the first heat source pipe **28**, and the first shutoff valve **32**, and the remaining is sent to the secondary flow path **35***a* of the cascade heat exchanger **35** via the secondary switching mechanism **22** and the third heat source pipe **25**.

The high-pressure refrigerant sent to the first connection pipe **8** is sent to the first branching pipe **63***c*. The high-pressure refrigerant sent to the first branching pipe **63***c* is sent to the utilization heat exchanger **52***c* in the utilization unit **3***c* via the first control valve **66***c* and the junction pipe **62***c*.

The high-pressure refrigerant sent to the utilization heat exchanger **52***c* exchanges heat with indoor air supplied by the indoor fan **53***c* in the utilization heat exchanger **52***c*. The refrigerant flowing in the utilization heat exchanger **52***c* thus radiates heat. Indoor air is heated and is supplied into the indoor space, and the utilization unit **3***c* executes heating operation. The refrigerant having radiated heat in the utilization heat exchanger **52***c* flows in the second utilization pipe **56***c* and is controlled in flow rate at the utilization expansion valve **51***c*. The refrigerant having flowed in the second connecting tube **16***c* is sent to the third branching pipe **61***c* in the branching unit **6***c*.

The refrigerant sent to the third branching pipe **61***c* is sent to the third connection pipe **7**.

The high-pressure refrigerant sent to the secondary flow path **35***a* of the cascade heat exchanger **35** exchanges heat with the primary refrigerant flowing in the primary flow path **35***b* in the cascade heat exchanger **35** to radiate heat. The secondary refrigerant having radiated heat in the cascade heat exchanger **35** is controlled in flow rate at the heat source expansion valve **36** and then flows into the receiver **45**. Part of the refrigerant having flowed out of the receiver **45** is branched into the subcooling circuit **48**, is decompressed at the subcooling expansion valve **48***a*, and then joins the suction flow path **23**. In the subcooling heat exchanger **47**, part of the remaining refrigerant having flowed out of the receiver **45** is cooled by the refrigerant flowing in the subcooling circuit **48**, is then sent to the third connection pipe **7** via the third shutoff valve **31**, and joins the refrigerant having radiated heat in the utilization heat exchanger **52***c*.

The refrigerant having joined in the third connection pipe **7** is branched into two portions to be sent to the third branching pipes **61***a* and **61***b* of the branching units **6***a* and **6***b*. Thereafter, the refrigerant having flowed in the second connecting tubes **16***a* and **16***b* is sent to the second utilization pipes **56***a* and **56***b* of the first and second utilization units **3***a* and **3***b*. The refrigerant flowing in the second utilization pipes **56***a* and **56***b* passes the utilization expansion valves **51***a* and **51***b* in the utilization units **3***a* and **3***b*.

The refrigerant having passed the utilization expansion valves **51***a* and **51***b* each controlled in opening degree exchanges heat with indoor air supplied by the indoor fans **53***a* and **53***b* in the utilization heat exchangers **52***a* and **52***b*. The refrigerant flowing in the utilization heat exchangers **52***a* and **52***b* is thus evaporated into a low-pressure gas refrigerant. Indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization heat exchangers **52***a* and **52***b* is sent to the junction pipes **62***a* and **62***b* of the first and second branching units **6***a* and **6***b*.

The low-pressure gas refrigerant sent to the junction pipes **62***a* and **62***b* is sent to the second connection pipe **9** via the second control valves **67***a* and **67***b* and the second branching pipes **64***a* and **64***b*, to join.

The low-pressure gas refrigerant sent to the second connection pipe **9** is returned to the suction side of the secondary compressor **21** via the second shutoff valve **33**, the second heat source pipe **29**, the suction flow path **23**, and the accumulator **30**.

Behavior during mainly cooling operation is executed in this manner.

(9-4) Mainly Heating Operation

During mainly heating operation, for example, the utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* each function as a refrigerant radiator, and the utilization heat exchanger **52***c* functions as a refrigerant evaporator. During mainly heating operation, the cascade heat exchanger **35** functions as an evaporator for the secondary refrigerant. During mainly heating operation, the primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** in the refrigeration cycle system **1** are configured as depicted in FIG. **6**. FIG. **6** includes arrows provided to the primary refrigerant circuit **5***a* and arrows provided to the secondary refrigerant circuit **10**, which indicate refrigerant flows during mainly heating operation.

Specifically, in the primary unit **5**, the primary switching mechanism **72** is switched into a sixth connection state to cause the cascade heat exchanger **35** to function as a radiator for the primary refrigerant. The sixth connection state of the primary switching mechanism **72** corresponds to a connection state depicted by broken lines in the primary switching mechanism **72** in FIG. **6**. Accordingly in the primary unit **5**, the primary refrigerant discharged from the primary compressor **71** passes the primary switching mechanism **72** and flows in the primary flow path **35***b* of the cascade heat exchanger **35** to be condensed. The primary refrigerant condensed in the cascade heat exchanger **35** is decompressed at the primary expansion valve **76**, then exchanges heat with outdoor air supplied from the primary fan **75** in the primary heat exchanger **74** to be evaporated, and is sucked into the primary compressor **71** via the primary switching mechanism **72**.

In the heat source unit **2**, the secondary switching mechanism **22** is switched into the second connection state as well as the third connection state. In the second connection state of the secondary switching mechanism **22**, the first switching valve **22***a* is in the closed state and the third switching valve **22***c* is in the opened state. In the third connection state of the secondary switching mechanism **22**, the second switching valve **22***b* is in the opened state and the fourth switching valve **22***d* is in the closed state. The cascade heat exchanger **35** thus functions as an evaporator for the secondary refrigerant. The heat source expansion valve **36** is controlled in opening degree. In the first to third branching units **6***a*, **6***b*, and **6***c*, the first control valves **66***a* and **66***b* and the second control valve **67***c* are controlled into the opened state, and the first control valve **66***c* and the second control valves **67***a* and **67***b* are controlled into the closed state. Accordingly, the utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* each function as a refrigerant radiator, and the utilization heat exchanger **52***c* in the utilization unit **3***c* functions as a refrigerant evaporator. The utilization heat exchanger **52***c* in the utilization unit **3***c* and the suction side of the secondary compressor **21** in the heat source unit **2** are connected via the first utilization pipe **57***c*, the first connecting tube **15***c*, the junction pipe **62***c*, the second branching pipe **64***c*, and the second connection pipe **9**. The utilization heat exchangers **52***a* and **52***b* in the utilization units **3***a* and **3***b* and the discharge side of the secondary compressor **21** in the heat source unit **2** are connected via the discharge flow path **24**, the first heat source pipe **28**, the first connection pipe **8**, the first branching pipes **63***a* and **63***b*, the junction pipes **62***a* and **62***b*, the first connecting tubes **15***a* and **15***b*, and the first utilization pipes **57***a* and **57***b*. The subcooling expansion valve **48***a* and the bypass expansion valve **46***a* are controlled into the closed state. In the utilization units **3***a*, **3***b*, and **3***c*, the utilization expansion valves **51***a*, **51***b*, and **51***c* are each controlled in opening degree.

In the secondary refrigerant circuit **10** in this state, a secondary high-pressure refrigerant compressed in and discharged from the secondary compressor **21** is sent to the first connection pipe **8** via the secondary switching mechanism **22**, the first heat source pipe **28**, and the first shutoff valve **32**.

The high-pressure refrigerant sent to the first connection pipe **8** is branched into two portions to be sent to the first branching pipes **63***a* and **63***b* of the first branching unit **6***a* and the second branching unit **6***b* respectively connected to the first utilization unit **3***a* and the second utilization unit **3***b* in operation. The high-pressure refrigerant sent to the first branching pipes **63***a* and **63***b* is sent to the utilization heat exchangers **52***a* and **52***b* in the first utilization unit **3***a* and the second utilization unit **3***b* via the first control valves **66***a* and **66***b*, the junction pipes **62***a* and **62***b*, and the first connecting tubes **15***a* and **15***b*.

The high-pressure refrigerant sent to the utilization heat exchangers **52***a* and **52***b* exchanges heat with indoor air supplied by the indoor fans **53***a* and **53***b* in the utilization heat exchangers **52***a* and **52***b*. The refrigerant flowing in the utilization heat exchangers **52***a* and **52***b* thus radiates heat. Indoor air is heated and is supplied into the indoor space. The indoor space is thus heated. The refrigerant having radiated heat in the utilization heat exchangers **52***a* and **52***b* flows in the second utilization pipes **56***a* and **56***b*, and passes the utilization expansion valves **51***a* and **51***b* each controlled in opening degree. Thereafter, the refrigerant having flowed in the second connecting tubes **16***a* and **16***b* is sent to the third connection pipe **7** via the third branching pipes **61***a* and **61***b* of the branching units **6***a* and **6***b*.

Part of the refrigerant sent to the third connection pipe **7** is sent to the third branching pipe **61***c* of the branching unit **6***c*, and the remaining is sent to the heat source expansion valve **36** via the third shutoff valve **31**.

The refrigerant sent to the third branching pipe **61***c* flows in the second utilization pipe **56***c* of the utilization unit **3***c* via the second connecting tube **16***c*, and is sent to the utilization expansion valve **51***c*.

The refrigerant having passed the utilization expansion valve **51***c* controlled in opening degree exchanges heat with indoor air supplied by the indoor fan **53***c* in the utilization heat exchanger **52***c*. The refrigerant flowing in the utilization heat exchanger **52***c* is thus evaporated into a low-pressure gas refrigerant. Indoor air is cooled and is supplied into the indoor space. The indoor space is thus cooled. The low-pressure gas refrigerant evaporated in the utilization heat exchanger **52***c* passes the first utilization pipe **57***c* and the first connecting tube **15***c* to be sent to the junction pipe **62***c*.

The low-pressure gas refrigerant sent to the junction pipe **62***c* is sent to the second connection pipe **9** via the second control valve **67***c* and the second branching pipe **64***c*.

The low-pressure gas refrigerant sent to the second connection pipe **9** is returned to the suction side of the secondary compressor **21** via the second shutoff valve **33**, the second heat source pipe **29**, the suction flow path **23**, and the accumulator **30**.

The refrigerant sent to the heat source expansion valve **36** passes the heat source expansion valve **36** controlled in opening degree, and then exchanges heat with the primary refrigerant flowing in the primary flow path **35***b* in the secondary flow path **35***a* of the cascade heat exchanger **35**. The refrigerant flowing in the secondary flow path **35***a* of the cascade heat exchanger **35** is evaporated into a low-pressure gas refrigerant and is sent to the secondary switching mechanism **22**. The low-pressure gas refrigerant sent to the secondary switching mechanism **22** joins the low-pressure gas refrigerant evaporated in the utilization heat exchanger

52*c* on the suction flow path **23**. The refrigerant thus joined is returned to the suction side of the secondary compressor **21** via the accumulator **30**.

Behavior during mainly heating operation is executed in this manner.

(10) Heat Storing Operation and Defrosting Operation

In the refrigeration cycle system **1**, heat storing operation and defrosting operation are executed when the predetermined condition is satisfied during normal operation of heating operation or mainly heating operation. Heat storing operation and defrosting operation will be described hereinafter with reference to a flowchart in FIG. **7**.

Described herein is a process flow from execution of heat storing operation and defrosting operation subsequent to heating operation or mainly heating operation, and then returning to heating operation or mainly heating operation.

In step S**1**, the control unit **80** controls various devices such that the refrigeration cycle system **1** executes normal operation of heating operation or mainly heating operation.

In step S**2**, the control unit **80** determines whether or not a predetermined defrosting condition is satisfied regarding adhesion of frost to the primary heat exchanger **74**. The defrosting condition is not limited, and determination can be made in accordance with at least one of conditions such as outdoor air temperature is equal to or less than a predetermined value, predetermined time has elapsed from completion of last defrosting operation, temperature of the primary heat exchanger **74** is equal to or less than a predetermined value, and evaporation pressure or evaporation temperature of the primary refrigerant is equal to or less than a predetermined value. The flow transitions to step S**3** if the defrosting condition is satisfied. Step S**2** is repeated if the defrosting condition is not satisfied.

In step S**3**, the control unit **80** starts first heat storing operation as the heat storing operation.

During first heat storing operation, the control unit **80** executes various control as follows. The refrigerant flow during first heat storing operation is similar to the refrigerant flow during heating operation depicted in FIG. **4**.

As to the primary refrigerant circuit **5***a*, the control unit **80** keeps the primary switching mechanism **72** in the connection state of normal operation, keeps the primary fan **75** in an operating state, and continuously drives the primary compressor **71**. The primary refrigerant thus flows in the order of the primary compressor **71**, the cascade heat exchanger **35**, the primary expansion valve **76**, and the primary heat exchanger **74**. The control unit **80** further controls a valve opening degree of the primary expansion valve **76** such that the refrigerant sucked into the primary compressor **71** has a degree of superheating at a predetermined value. Alternatively, the control unit **80** may control to increase a drive frequency of the primary compressor **71** so as to be higher than the drive frequency during normal operation, or may control the drive frequency of the primary compressor **71** to a predetermined maximum frequency.

As to the secondary refrigerant circuit **10**, the control unit **80** stops the indoor fans **53***a*, **53***b*, and **53***c*. Upon transition from heating operation to first heat storing operation, the control unit **80** keeps the connection state of the secondary switching mechanism **22**, keeps the utilization expansion valves **51***a*, **51***b*, and **51***c* and the first control valves **66***a*, **66***b*, and **66***c* in the opened state, and keeps the second control valves **67***a*, **67***b*, and **67***c*, the subcooling expansion valve **48***a*, and the bypass expansion valve **46***a* in the closed state. Upon transition from mainly heating operation to first heat storing operation, the control unit **80** keeps the connection state of the secondary switching mechanism **22**, controls the utilization expansion valves **51***a*, **51***b*, and **51***c* and the first control valves **66***a*, **66***b*, and **66***c* into the opened state, and controls the second control valves **67***a*, **67***b*, and **67***c*, the subcooling expansion valve **48***a*, and the bypass expansion valve **46***a* into the closed state. The secondary refrigerant thus flows in the order of the secondary compressor **21**, the utilization heat exchangers **52***a*, **52***b*, and **52***c*, the utilization expansion valves **51***a*, **51***b*, and **51***c*, and the cascade heat exchanger **35**. The control unit **80** controls a valve opening degree of the heat source expansion valve **36** such that the refrigerant sucked into the secondary compressor **21** has a degree of superheating at a predetermined value. Alternatively, the secondary compressor **21** may keep driven, or may be controlled to have a higher drive frequency in comparison to normal operation.

In step S**4**, the control unit **80** determines whether or not a first heat storage completion condition is satisfied. The first heat storage completion condition is not limited in this case, and determination can be made in accordance with at least one of conditions such as predetermined time has elapsed from the start of first heat storing operation, the cascade heat exchanger **35** has temperature equal to or more than a predetermined value, the secondary refrigerant discharged from the secondary compressor **21** has pressure equal to or more than a predetermined value, the secondary refrigerant discharged from the secondary compressor **21** has temperature equal to or more than a predetermined value, and the secondary refrigerant at a predetermined site having a flow of a liquid refrigerant on the secondary refrigerant circuit **10** has temperature equal to or more than a predetermined value. The flow transitions to step S**5** if the first heat storage completion condition is satisfied. Step S**3** is repeated if the first heat storage completion condition is not satisfied.

In step S**5**, the control unit **80** ends first heat storing operation, switches the secondary switching mechanism **22** into the first connection state as well as the fourth connection state after executing pressure equalizing behavior at the secondary refrigerant circuit **10**, controls the utilization expansion valves **51***a*, **51***b*, and **51***c* into the closed state, and starts second heat storing operation as heat storing operation. Alternatively, the first control valves **66***a*, **66***b*, and **66***c* and the second control valves **67***a*, **67***b*, and **67***c* may be controlled into the closed state in this case.

During second heat storing operation, the control unit **80** executes various control as follows. FIG. **8** indicates a refrigerant flow during second heat storing operation.

As to the primary refrigerant circuit **5***a*, the control unit **80** keeps operation similar to first heat storing operation.

As to the secondary refrigerant circuit **10**, the control unit **80** switches the secondary switching mechanism **22** into the first connection state as well as the fourth connection state with the indoor fans **53***a*, **53***b*, and **53***c* being kept stopped, and drives the secondary compressor **21** while controlling the utilization expansion valves **51***a*, **51***b*, and **51***c*, the first control valves **66***a*, **66***b*, and **66***c*, the second control valves **67***a*, **67***b*, and **67***c*, and the subcooling expansion valve **48***a* into the closed state and controlling the bypass expansion valve **46***a* into the opened state. The secondary refrigerant thus flows in the order of the secondary compressor **21**, the cascade heat exchanger **35**, the receiver **45**, the bypass circuit **46**, and the bypass expansion valve **46***a*. The heat source expansion valve **36** is controlled into a fully opened state. The control unit **80** controls the drive frequency such that a high pressure refrigerant and a low pressure refrigerant at the secondary refrigerant circuit **10** have differential pressure secured to be equal to or more than a predetermined value in the secondary compressor **21**. The control unit **80** controls a valve opening degree of the bypass expansion valve **46***a* in accordance with temperature of the cascade heat exchanger **35** and a degree of superheating of the refrigerant discharged from the secondary compressor **21**. Specifically, the control unit **80** controls to increase the valve opening degree such that the secondary refrigerant has a secured flow in the cascade heat exchanger **35** and the cascade heat exchanger **35** has temperature kept equal to or more than a predetermined value, and controls to decrease the valve opening degree such that the refrigerant discharged from the secondary compressor **21** has a degree of superheating kept equal to or more than a predetermined value to prevent a wet state of the secondary refrigerant sucked into the secondary compressor **21**, so as to control the valve opening degree of the bypass expansion valve **46***a*.

In step S**6**, the control unit **80** determines whether or not a second heat storage completion condition is satisfied. The second heat storage completion condition is not limited in this case, and determination can be made in accordance with at least one of conditions such as predetermined time has elapsed from the start of second heat storing operation, the secondary refrigerant discharged from the secondary compressor **21** has pressure equal to or more than a predetermined value, the secondary refrigerant discharged from the secondary compressor **21** has temperature equal to or more than a predetermined value, the primary refrigerant discharged from the primary compressor **71** has pressure equal to or more than a predetermined value, the primary refrigerant discharged from the primary compressor **71** has temperature equal to or more than a predetermined value, and the cascade heat exchanger **35** has temperature equal to or more than a predetermined value. The control unit **80** may alternatively determine that the second heat storage completion condition is satisfied if the primary control unit **70** configured to control the primary refrigerant circuit **5***a* determines completion of preparation for starting defrosting operation at the primary refrigerant circuit **5***a*. The flow transitions to step S7 if the second heat storage completion condition is satisfied. Step S**5** is repeated if the second heat storage completion condition is not satisfied.

In step S7, the control unit **80** ends second heat storing operation and starts defrosting operation.

During defrosting operation, the control unit **80** executes various control as follows. FIG. **9** indicates a refrigerant flow during defrosting operation.

As to the primary refrigerant circuit **5***a*, the control unit **80** switches the primary switching mechanism **72** into the fifth connection state after executing pressure equalizing behavior at the primary refrigerant circuit **5***a*, and drives the primary compressor **71** while keeping the primary fan **75** stopped. The primary refrigerant thus flows in the order of the primary compressor **71**, the primary heat exchanger **74**, the primary expansion valve **76**, and the cascade heat exchanger **35**. The control unit **80** controls the valve opening degree of the primary expansion valve **76** such that the degree of superheating of the refrigerant sucked into the primary compressor **71** is kept at the predetermined value. Alternatively, the control unit **80** may control to increase a drive frequency of the primary compressor **71** so as to be higher than the drive frequency during normal operation, or may control the drive frequency of the primary compressor **71** to a predetermined maximum frequency.

As to the secondary refrigerant circuit **10**, the control unit **80** keeps control for second heat storing operation.

In step S**8**, the control unit **80** determines whether or not a defrosting completion condition is satisfied. The defrosting completion condition is not limited, and determination can be made in accordance with at least one of conditions such as predetermined time has elapsed from the start of defrosting operation, the primary heat exchanger **74** has temperature equal to or more than a predetermined value, and condensation pressure or condensation temperature of the primary refrigerant is equal to or more than a predetermined value. The flow transitions to step S**9** if the defrosting completion condition is satisfied. Step S7 is repeated if the defrosting completion condition is not satisfied.

In step S9, the control unit **80** controls various devices such that the refrigeration cycle system **1** returns to heating operation or mainly heating operation.

(11) Characteristics of Embodiment

The refrigeration cycle system **1** according to the present embodiment executes first heat storing operation and second heat storing operation as heat storing operation before starting defrosting operation.

During first heat storing operation, the secondary compressor **21** is driven with the indoor fans **53***a*, **53***b*, and **53***c* being stopped in the secondary refrigerant circuit **10**. This inhibits heat release from the secondary refrigerant in the utilization heat exchangers **52***a*, **52***b*, and **52***c*, and enables heat storage in the secondary refrigerant circuit **10**. In particular, the indoor fans **53***a*, **53***b*, and **53***c* are stopped, so that the secondary refrigerant having passed the utilization heat exchangers **52***a*, **52***b*, and **52***c*, with heat release being inhibited, reaches the secondary flow path **35***a* of the cascade heat exchanger **35** to enable heat storage in the cascade heat exchanger **35**.

Furthermore, during second heat storing operation, the secondary refrigerant circuit **10** has circulation such that the bypass expansion valve **46***a* is opened to cause the secondary refrigerant to flow to the bypass circuit **46**, with the utilization expansion valves **51***a*, **51***b*, and **51***c* being closed to stop supply of the secondary refrigerant to the utilization circuits **13***a*, **13***b*, and **13***c*. Accordingly, the utilization heat exchangers **52***a*, **52***b*, and **52***c* have inhibited temperature decrease while a high-temperature high-pressure refrigerant discharged from the secondary compressor **21** is supplied to the secondary flow path **35***a* of the cascade heat exchanger **35** to cause the cascade heat exchanger **35** to store heat, so as to inhibit deterioration of a utilization environment.

In the primary refrigerant circuit **5***a* during first heat storing operation and second heat storing operation, the high-temperature high-pressure refrigerant discharged from the primary compressor **71** is sent to the primary flow path **35***b* of the cascade heat exchanger **35**. This also promotes heat storage at the cascade heat exchanger **35**.

The refrigeration cycle system **1** according to the present embodiment accordingly achieves sufficient heat storage for melting frost at the primary heat exchanger **74** during defrosting operation prior to execution of the defrosting operation.

In the secondary refrigerant circuit **10** during defrosting operation, the cascade heat exchanger **35** can be supplied with heat by sending the high-temperature high-pressure refrigerant discharged from the secondary compressor **21** to the secondary flow path **35***a* of the cascade heat exchanger **35** while stopping supply of the secondary refrigerant to the utilization circuits **13***a*, **13***b*, and **13***c*. In the primary refrigerant circuit **5***a*, the primary refrigerant flowing in the primary flow path **35***b* of the cascade heat exchanger **35** can be provided with heat supplied from the secondary refrigerant to the cascade heat exchanger **35**, and the primary compressor **71** can further pressurize the primary refrigerant thus provided with the heat in order to melt frost at the primary heat exchanger **74** by means of the refrigerant brought into the high-temperature high-pressure state. Frost at the primary heat exchanger **74** can thus be molten efficiently. This shortens time of deterioration of the utilization environment due to execution of defrosting operation.

During second heat storing operation and defrosting operation described above, the secondary refrigerant flows to the bypass circuit **46** extending from the gas phase region in the receiver **45** on the secondary refrigerant circuit **10**. This allows the secondary refrigerant flowing in the bypass circuit **46** to principally become a gas refrigerant, so as to easily inhibit the refrigerant sucked into the secondary compressor **21** from coming into the wet state.

During second heat storing operation and defrosting operation described above, the valve opening degree of the bypass expansion valve **46***a* is controlled such that the cascade heat exchanger **35** has temperature kept equal to or more than a predetermined value and the refrigerant discharged from the secondary compressor **21** has a degree of superheating kept equal to or more than a predetermined value. Assuming that the secondary refrigerant has a stopped flow in the secondary flow path **35***a* of the cascade heat exchanger **35**, the primary refrigerant evaporates at the primary flow path **35***b* of the cascade heat exchanger **35**, so that heat keeps released from the secondary refrigerant stopped in the secondary flow path **35***a*. Accordingly, the secondary refrigerant is decreased in temperature in the secondary flow path **35***a*, and the cascade heat exchanger **35** is also decreased in temperature, to decrease heat used for melting frost at the primary heat exchanger **74** during defrosting operation. In contrast, the valve opening degree of the bypass expansion valve **46***a* is controlled such that temperature of the cascade heat exchanger **35** is kept equal to or more than a predetermined value, so as to inhibit the secondary refrigerant from being stopped in the secondary flow path **35***a* and secure sufficient heat for defrosting operation. Furthermore, the valve opening degree of the bypass expansion valve **46***a* is controlled to inhibit the wet state of the secondary refrigerant sucked into the secondary compressor **21**. This sufficiently secures heat for defrosting operation and inhibits liquid compression in the secondary compressor **21**.

In the refrigeration cycle system **1** according to the present embodiment, adoption of carbon dioxide as a refrigerant in the secondary refrigerant circuit **10** decreases the global warming potential (GWP). Even if the refrigerant containing no chlorofluorocarbon leaks on the utilization side, there is no outflow of chlorofluorocarbon on the utilization side.

The refrigeration cycle system **1** according to the present embodiment adopts the binary refrigeration cycle, to exhibit sufficient capacity at the secondary refrigerant circuit **10**.

(12) Other Embodiments

(12-1) Other Embodiment A

The above embodiment exemplifies execution of first heat storing operation and second heat storing operation as heat storing operation before starting defrosting operation.

For example, heat storing operation executed before the start of defrosting operation may alternatively include only first heat storing operation or only second heat storing operation.

When only first heat storing operation is executed as heat storing operation, defrosting operation according to the above embodiment may start upon satisfaction of the first heat storage completion condition. Alternatively, control of defrosting operation may start at the primary refrigerant circuit **5***a* after first heat storing operation ends and control of defrosting operation starts at the secondary refrigerant circuit **10**. In other words, control of defrosting operation at the primary refrigerant circuit **5***a* may not start before control of defrosting operation at the secondary refrigerant circuit **10**. In this case, in an exemplary case where the first heat storage completion condition is satisfied, the secondary switching mechanism **22** may be switched into the first connection state as well as the fourth connection state in the secondary refrigerant circuit **10**, and the primary compressor **71** on the primary refrigerant circuit **5***a* may be stopped until the primary control unit **70** determines that preparation for the start of defrosting operation completes in the primary refrigerant circuit **5***a*. When the primary control unit **70** thereafter determines that preparation for the start of defrosting operation completes in the primary refrigerant circuit **5***a*, the secondary compressor **21** on the secondary refrigerant circuit **10** may be activated and the primary compressor **71** on the primary refrigerant circuit **5***a* may subsequently be activated. Pressure equalizing behavior at the primary refrigerant circuit **5***a* and switching the primary switching mechanism **72** into the fifth connection state may be executed after satisfaction of the first heat storage completion condition, or upon determination by the primary control unit **70** that preparation for the start of defrosting operation is completed in the primary refrigerant circuit **5***a*. The above control allows the primary flow path **35***b* of the cascade heat exchanger **35** to function as an evaporator for the primary refrigerant and allows the secondary flow path **35***a* to function as an evaporator for the secondary refrigerant, to avoid a situation where the primary refrigerant flowing in the primary flow path **35***b* becomes less likely to gain heat from the secondary refrigerant flowing in the secondary flow path **35***a*.

When only second heat storing operation is executed as heat storing operation, second heat storing operation starts upon satisfaction of the defrosting condition, and defrosting operation thereafter starts upon satisfaction of the second heat storage completion condition. Alternatively, upon satisfaction of the second heat storage completion condition, similarly to the above case, control of defrosting operation may start in the primary refrigerant circuit **5***a* after control of defrosting operation initially starts in the secondary refrigerant circuit **10**. In an exemplary case where the second heat storage completion condition is satisfied, with the secondary switching mechanism **22** being kept in its connection state in the secondary refrigerant circuit **10**, the primary compressor **71** on the primary refrigerant circuit **5***a* may be stopped until the primary control unit **70** determines that preparation for the start of defrosting operation completes in the primary refrigerant circuit **5***a*. When the primary control unit **70** thereafter determines that preparation for the start of defrosting operation completes in the primary refrigerant circuit **5***a*, the secondary compressor **21** on the secondary refrigerant circuit **10** may be activated and the primary compressor **71** on the primary refrigerant circuit **5***a* may subsequently be activated. Pressure equalizing behavior at the primary refrigerant circuit **5***a* and switching the primary switching mechanism **72** into the fifth connection state may be executed after satisfaction of the second heat storage completion condition, or upon determination by the primary control unit **70** that preparation for the start of defrosting operation completes in the primary refrigerant circuit **5***a*. Similarly to the above case, it is possible to avoid the situation where the primary refrigerant flowing in the primary flow path **35***b* is less likely to gain heat from the secondary refrigerant flowing in the secondary flow path **35***a*.

(12-2) Other Embodiment B

The above embodiment exemplifies the case where a refrigerant flows in the bypass circuit **46** during second heat storing operation and defrosting operation.

The bypass circuit **46** extends from the gas phase region in the receiver **45**. It is thus possible to send a gas-phase refrigerant toward the suction side of the secondary compressor **21** until the receiver **45** is filled with a refrigerant in the liquid state.

In an exemplary case where second heat storing operation or defrosting operation is continuously executed to satisfy a full liquid condition as to the receiver **45** being filled with a liquid refrigerant, the subcooling expansion valve **48***a* may be opened instead of opening the bypass expansion valve **46***a* or along with opening the bypass expansion valve **46***a*, so as to cause the refrigerant to flow also in the subcooling circuit **48**.

The full liquid condition as to the receiver **45** being filled with a liquid refrigerant may be exemplarily determined in accordance with the degree of superheating of the refrigerant flowing downstream of the bypass expansion valve **46***a* in the bypass circuit **46**. The degree of superheating may be obtained from temperature detected by the bypass circuit temperature sensor **85**, pressure detected by the secondary suction pressure sensor **37**, and the like.

(12-3) Other Embodiment C

The above embodiment exemplifies the case where the indoor fans **53***a*, **53***b*, and **53***c* are stopped during first heat storing operation.

However, first heat storing operation is not limited to control to stop the indoor fans **53***a*, **53***b*, and **53***c* during first heat storing operation. For example, the indoor fans **53***a*, **53***b*, and **53***c* may alternatively be controlled to have airflow volume less than airflow volume during normal operation of heating operation or mainly heating operation. This case also inhibits heat release from the secondary refrigerant in the utilization heat exchangers **52***a*, **52***b*, and **52***c*.

(12-4) Other Embodiment D

The above embodiment exemplifies the case where the utilization expansion valves **51***a*, **51***b*, and **51***c* are controlled into the closed state during second heat storing operation and defrosting operation.

However, second heat storing operation and defrosting operation are not limited to control to completely stop the utilization expansion valves **51***a*, **51***b*, and **51***c*. For example, the utilization expansion valves **51***a*, **51***b*, and **51***c* may alternatively be controlled to be smaller in valve opening degree in comparison to normal operation of heating operation or mainly heating operation. This case also inhibits quantity of the secondary refrigerant sent to the utilization heat exchangers **52***a*, **52***b*, and **52***c*, so as to inhibit heat release at the utilization heat exchangers **52***a*, **52***b*, and **52***c*.

Control to open, without closing, the utilization expansion valves **51***a*, **51***b*, and **51***c* may not be executed during heat storing operation or until satisfaction of the predetermined condition from the start of defrosting operation, but may be executed after the start of defrosting operation and upon satisfaction of the following condition. Specifically, examples of the condition include a case where the secondary refrigerant sucked into the secondary compressor **21** in the secondary refrigerant circuit **10** has a degree of superheating equal to or less than a predetermined value, a case where the secondary refrigerant discharged from the secondary compressor **21** has a degree of superheating equal to or less than a predetermined value, a case where the secondary refrigerant in the secondary refrigerant circuit **10** has high pressure equal to or less than a predetermined value, a case where the liquid refrigerant in the secondary refrigerant circuit **10** has temperature equal to or less than a predetermined value, and a case where defrosting operation cannot end even after elapse of predetermined time from the start of defrosting operation.

Whether or not the secondary refrigerant in the secondary refrigerant circuit **10** has high pressure equal to or less than the predetermined value may be exemplarily determined in accordance with pressure detected by the secondary discharge pressure sensor **38**. Whether or not the liquid refrigerant in the secondary refrigerant circuit **10** has temperature equal to or less than the predetermined value may be determined in accordance with temperature detected by the receiver outlet temperature sensor **84**, temperature detected by the subcooling outlet temperature sensor **86**, or the like.

(12-5) Other Embodiment E

The above embodiment exemplifies control to supply the primary flow path **35***b* of the cascade heat exchanger **35** with the refrigerant discharged from the primary compressor **71** in the primary refrigerant circuit **5***a* during heat storing operation.

Alternatively, the primary compressor **71** may be stopped during heat storing operation. Furthermore, pressure equalizing control and control to switch the primary switching mechanism **72** may be completed to achieve the connection state enabling the start of defrosting operation, and the primary compressor **71** may be made standby to be activated until completion of heat storing operation.

Alternatively, the primary compressor **71** may be driven as in the above embodiment during first heat storing operation as heat storing operation, and the primary compressor **71** may be stopped during second heat storing operation, pressure equalizing control and control to switch the primary switching mechanism **72** may be completed, and the primary compressor **71** may be made standby to be activated until completion of heat storing operation. This avoids both the primary flow path **35***b* and the secondary flow path **35***a* of the cascade heat exchanger **35** from functioning as a refrigerant radiator during second heat storing operation, to inhibit abnormal increase of high pressure of the primary refrigerant or the secondary refrigerant.

(12-6) Other Embodiment F

The above embodiment exemplifies the case where the bypass expansion valve **46***a* is controlled to open so as to cause the secondary refrigerant to flow in the bypass circuit **46** during second heat storing operation and defrosting operation.

Alternatively, instead of causing the secondary refrigerant to flow to the bypass circuit **46** during second heat storing operation and defrosting operation, the subcooling expansion valve **48***a* may be controlled to open so as to cause the secondary refrigerant to flow to the subcooling circuit **48**.

(12-7) Other Embodiment G

The above embodiment exemplifies R32 as the refrigerant provided in the primary refrigerant circuit **5***a* and carbon dioxide as the refrigerant provided in the secondary refrigerant circuit **10**.

However, the refrigerant provided in the primary refrigerant circuit **5***a* should not be limited, and examples thereof include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

Furthermore, the refrigerant provided in the secondary refrigerant circuit **10** should not be limited, and examples thereof include HFC-32, an HFO refrigerant, a refrigerant obtained by mixing HFC-32 and the HFO refrigerant, carbon dioxide, ammonia, and propane.

Examples of the HFO refrigerant include HFO-1234yf and HFO-1234ze.

The primary refrigerant circuit **5***a* and the secondary refrigerant circuit **10** may adopt a same refrigerant or different refrigerants.

(12-8) Other Embodiment H

The above embodiment exemplifies, as the secondary refrigerant circuit **10**, a refrigerant circuit having three pipes of the first connection pipe **8**, the second connection pipe **9**, and the third connection pipe **7**, and configured to simultaneously execute cooling operation and heating operation.

However, the secondary refrigerant circuit **10** should not be limited to such a refrigerant circuit configured to simultaneously execute cooling operation and heating operation, and may be a circuit including the heat source unit **2** and the utilization units **3***a*, **3***b*, and **3***c* connected via two connection pipes.

(12-9) Other Embodiment I

The above embodiment exemplifies execution of first heat storing operation and second heat storing operation as heat storing operation before starting defrosting operation.

Alternatively, defrosting operation may start earlier upon satisfaction of the defrosting condition without execution of heat storing operation described above before the start of defrosting operation.

(12-10) Others

The cascade heat exchanger may be configured to cause heat exchange between the first refrigerant and the second refrigerant.

The refrigeration cycle system may include a control unit configured to execute the defrosting operation.

During the defrosting operation, the second refrigerant having passed the cascade heat exchanger may entirely flow into the bypass circuit, or the second refrigerant having passed the cascade heat exchanger may partially flow into the bypass circuit.

The portion between the utilization heat exchanger and the cascade heat exchanger may have a flow of the second refrigerant having high pressure or intermediate pressure during operation of causing the second refrigerant to flow from the second compressor toward the cascade heat exchanger.

During the defrosting operation, the controlling valve may be opened constantly or at least temporarily.

The controlling valve may be configured to be switchable between an opened state and a closed state, or may be configured to have a controllable valve opening degree.

In an exemplary case where the second circuit includes an accumulator provided downstream of the second compressor, the suction flow path of the second compressor can be a pipe including a flow path from the second switching unit to the accumulator and a flow path from the accumulator to the second compressor.

The opening degree of the expansion valve before the defrosting operation starts is not particularly limited, and can be an opening degree of the expansion valve during normal operation in the heating cycle, and the opening degree may be controlled in accordance with an operation condition. The opening degree may exemplarily be controlled in accordance with a degree of superheating of the second refrigerant sucked into the second compressor or a degree of superheating of the second refrigerant discharged from the second compressor.

Examples of the predetermined condition include a case where the degree of superheating of the second refrigerant sucked into the second compressor is equal to or less than a predetermined value, a case where the degree of superheating of the second refrigerant discharged from the second compressor is equal to or less than a predetermined value, a case where the pressure of the high pressure refrigerant in the refrigeration cycle of the second circuit is equal to or less than a predetermined value, and a case where the temperature of the second refrigerant flowing between the utilization heat exchanger and the cascade heat exchanger on the second circuit is equal to or less than a predetermined value.

Being downstream of the portion connected with the bypass circuit indicates being downstream in a flow direction of the second refrigerant in the suction flow path.

The refrigerant cooler may be configured to cool the refrigerant having passed the cascade heat exchanger and flowing toward the utilization heat exchanger.

When the second circuit includes the receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, there may be provided two bypass circuits, namely, a bypass circuit passing the refrigerant cooler and a bypass circuit guiding the gas refrigerant in the receiver to the suction flow path of the second compressor. Furthermore, when the second circuit includes the receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, the bypass circuit may cause the gas refrigerant in the receiver to pass the refrigerant cooler and then guide the refrigerant to the suction flow path of the second compressor.

Appendix

The embodiments of the present disclosure have been described above. Various modifications to modes and details should be available without departing from the object and the scope of the present disclosure recited in the patent claims.

REFERENCE SIGNS LIST

1: refrigeration cycle system
2: heat source unit
3*a*: first utilization unit
3*b*: second utilization unit
3*c*: third utilization unit
4: secondary unit
5: primary unit
5*a*: primary refrigerant circuit (first circuit)
7: liquid-refrigerant connection pipe
8: high and low-pressure gas-refrigerant connection pipe
9: low-pressure gas-refrigerant connection pipe
10: secondary refrigerant circuit (second circuit)
11: heat source expansion mechanism
12: heat source circuit
13*a-c*: utilization circuit
20: heat source control unit
21: secondary compressor (second compressor)
21*a*: compressor motor
22: secondary switching mechanism (second switching unit)
23: suction flow path
24: discharge flow path
25: third heat source pipe
26: fourth heat source pipe
27: fifth heat source pipe
28: first heat source pipe
29: second heat source pipe
30: accumulator
31: third shutoff valve
32: first shutoff valve
33: second shutoff valve
34: oil separator
35: cascade heat exchanger
35*a*: secondary flow path
35*b*: primary flow path
36: heat source expansion valve
37: secondary suction pressure sensor
38: secondary discharge pressure sensor
39: secondary discharge temperature sensor
40: oil return circuit
41: oil return flow path
42: oil return capillary tube
44: oil return on-off valve
45: receiver
46: bypass circuit (bypass circuit)
46*a*: bypass expansion valve (controlling valve)
47: subcooling heat exchanger (refrigerant cooler)
48: subcooling circuit (bypass circuit)
48*a*: subcooling expansion valve (controlling valve)
50*a-c*: utilization control unit
51*a-c*: utilization expansion valve (expansion valve)
52*a-c*: utilization heat exchanger (utilizing heat exchanger)
53*a-c*: indoor fan
56*a*, **56***b*, **56***c*: second utilization pipe
57*a*, **57***b*, **57***c*: first utilization pipe
58*a*, **58***b*, **58***c*: liquid-side temperature sensor
60*a*, **60***b*, **60***c*: branching unit control unit
61*a*, **61***b*, **61***c*: third branching pipe
62*a*, **62***b*, **62***c*: junction pipe
63*a*, **63***b*, **63***c*: first branching pipe
64*a*, **64***b*, **64***c*: second branching pipe
66*a*, **66***b*, **66***c*: first control valve
67*a*, **67***b*, **67***c*: second control valve
70: primary control unit
71: primary compressor (first compressor)
72: primary switching mechanism (first switching unit)
74: primary heat exchanger (heat source heat exchanger)
76: primary expansion valve
77: outdoor air temperature sensor
78: primary discharge pressure sensor
79: primary suction pressure sensor
81: primary suction temperature sensor
82: primary heat-exchange temperature sensor
83: secondary cascade temperature sensor
84: receiver outlet temperature sensor
85: bypass circuit temperature sensor
86: subcooling outlet temperature sensor
87: subcooling circuit temperature sensor
88: secondary suction temperature sensor
80: control unit

CITATION LIST

Patent Literature

Patent Literature 1: Japanese Laid-Open Patent Publication No. 2014-109405

The invention claimed is:

1. A refrigeration cycle system comprising:

a first circuit configured to circulate a first refrigerant and including a first compressor, a cascade heat exchanger, a heat source heat exchanger, and first switching valves configured to switch a flow path of the first refrigerant;

a second circuit configured to circulate a second refrigerant and including a second compressor, the cascade heat exchanger, a utilization heat exchanger; and second switching valves configured to switch a flow path of the second refrigerant; wherein

the second circuit includes a bypass circuit connecting a portion between the utilization heat exchanger and the cascade heat exchanger and a suction flow path of the second compressor, and a controlling valve provided on the bypass circuit; and

an electronic controller configured to control the system to executes defrosting operation of circulating the first refrigerant in the order of the first compressor, the heat source heat exchanger, and the cascade heat exchanger, and circulating the second refrigerant in the order of the second compressor, the cascade heat exchanger, and the bypass circuit.

2. The refrigeration cycle system according to claim **1**, wherein

the second circuit includes an expansion valve provided between the utilization heat exchanger and a site branching to the bypass circuit in the portion between the utilization heat exchanger and the cascade heat exchanger.

3. The refrigeration cycle system according to claim **2**, wherein

the electronic controller is configured to control the expansion valve during the defrosting operation to be smaller in opening degree than the expansion valve before the defrosting operation starts.

4. The refrigeration cycle system according to claim **2**, wherein

the electronic controller is configured to control the expansion valve to be in a closed state during the defrosting operation.

5. The refrigeration cycle system according to claim **2**, wherein the electronic controller is configured to control, during the defrosting operation, the controlling valve to be decreased in opening degree and the expansion valve is increased in opening degree when at least one of a degree of superheating of the second refrigerant sucked into the second compressor, a degree of superheating of the second refrigerant discharged from the second compressor, pressure of a high pressure refrigerant in a refrigeration cycle of the second circuit, temperature of the second refrigerant flowing between the utilization heat exchanger and the cascade heat exchanger on the second circuit, and time elapsed from a start of the defrosting operation, satisfies a predetermined condition.

6. The refrigeration cycle system according to claim **1**, wherein

the second circuit includes an accumulator provided downstream of a portion connected with the bypass circuit on the suction flow path of the second compressor.

7. The refrigeration cycle system according to claim **1**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

8. The refrigeration cycle system according to claim **1**, wherein

the second circuit includes a refrigerant cooler provided between the cascade heat exchanger and the utilization heat exchanger, and

the bypass circuit passes the refrigerant cooler.

9. The refrigeration cycle system according to claim **3**, wherein

the electronic controller is configured to control, during the defrosting operation, the controlling valve to be decreased in opening degree and the expansion valve is increased in opening degree when at least one of a degree of superheating of the second refrigerant sucked into the second compressor, a degree of superheating of the second refrigerant discharged from the second compressor, pressure of a high pressure refrigerant in a refrigeration cycle of the second circuit, temperature of the second refrigerant flowing between the utilization heat exchanger and the cascade heat exchanger on the second circuit, and time elapsed from a start of the defrosting operation, satisfies a predetermined condition.

10. The refrigeration cycle system according to claim **4**, wherein

the electronic controller is configured to control, during the defrosting operation, the controlling valve to be decreased in opening degree and the expansion valve is increased in opening degree when at least one of a degree of superheating of the second refrigerant sucked into the second compressor, a degree of superheating of the second refrigerant discharged from the second compressor, pressure of a high pressure refrigerant in a refrigeration cycle of the second circuit, temperature of the second refrigerant flowing between the utilization heat exchanger and the cascade heat exchanger on the second circuit, and time elapsed from a start of the defrosting operation, satisfies a predetermined condition.

11. The refrigeration cycle system according to claim **2**, wherein

the second circuit includes an accumulator provided downstream of a portion connected with the bypass circuit on the suction flow path of the second compressor.

12. The refrigeration cycle system according to claim **3**, wherein

the second circuit includes an accumulator provided downstream of a portion connected with the bypass circuit on the suction flow path of the second compressor.

13. The refrigeration cycle system according to claim **4**, wherein

the second circuit includes an accumulator provided downstream of a portion connected with the bypass circuit on the suction flow path of the second compressor.

14. The refrigeration cycle system according to claim **5**, wherein

the second circuit includes an accumulator provided downstream of a portion connected with the bypass circuit on the suction flow path of the second compressor.

15. The refrigeration cycle system according to claim **2**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

16. The refrigeration cycle system according to claim **3**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

17. The refrigeration cycle system according to claim **4**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

18. The refrigeration cycle system according to claim **5**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

19. The refrigeration cycle system according to claim **6**, wherein

the second circuit includes a receiver provided between the cascade heat exchanger and the utilization heat exchanger and configured to reserve the second refrigerant, and

the bypass circuit guides a gas refrigerant in the receiver to the suction flow path of the second compressor.

20. The refrigeration cycle system according to claim **2**, wherein the second circuit includes a refrigerant cooler provided between the cascade heat exchanger and the utilization heat exchanger, and

the bypass circuit passes the refrigerant cooler.

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